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A NEW BAYESIAN ECOLOGICAL INFERENCE MODEL WITH COVARIATES

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Abstract

Ecological inference models are essential tools for estimating transition matrices from aggregate data, particularly in fields like political science and epidemiology. Building on the multinomial-Dirichlet framework developed by King, Rosen, Tanner, and Jiang (2001), this paper proposes a refined ecological inference model that introduces an alternative covariate formulation. Our approach maintains the flexibility and relaxed assumptions of the original model while addressing key structural and computational limitations, such as numerical stability and efficiency.

We further investigate the role of random effects by implementing two model variants, with and without random effects, to evaluate their impact on interpretability, computational speed, convergence, and robustness to unobserved heterogeneity. Using real-world voter registration data from Louisiana and simulated data with New Zealand elections as reference, we benchmark the proposed model against established Bayesian and numerical ecological inference methods, including the original covariate model.

Our results demonstrate that the refined model consistently matches or exceeds the accuracy of existing approaches while reducing computational time by up to 80%, and achieving greater numerical stability. Notably, excluding random effects can lead to order-of-magnitude speed improvements without significantly compromising accuracy at the global level. We also highlight the importance of covariate relevance, finding that only covariates strongly correlated with the latent transition structure substantially improve estimation. These findings clarify when model complexity is justified and when simpler models suffice.

In summary, this study contributes to the field of ecological inference by introducing a robust, data-supported, and computationally streamlined method for integrating covariates into the estimation of transition matrices. We explore the wider relevance of our approach across different domains and suggest future research avenues, such as developing faster numerical techniques and refining prior assumptions for covariate modelling.

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1 Introduction

Ecological inference is the process of making inferences about the behaviour of individuals given the observed aggregated of the population. This is achieved by estimating the unobserved link between two categorical groupings of a population. The link is defined as the proportion of people in a given category of the first or initial grouping that end up in a specific category of the second or outcome grouping. This problem is equivalent to finding the table that produced some marginals, with the rows normally corresponding to the initial groupings and the columns to the outcome groupings. For example, if someone knows how many people are in each age group, and the number of votes all political parties received in a specific election for multiple regions, one might be interested in how much support the different age groups gave to each of the parties. Using ecological inference would then give us the table with the percentage of support each age group gave to the different parties. This table, formed by all pairs of age groups and political parties, with each cell indicating the percentage or probability that someone of that age group would vote that party, is what is known as the transition matrix.

Unfortunately, a direct approach is not possible because the aggregate data can be explained by multiple distinct individual patterns. More formally, we want to find the transition matrix, that describes the change between a group with $R \geq 2$ categories (the number of rows of the matrix) and another group with $C \geq 2$ categories (the number of columns of the matrix). Then, the number of parameters that need to be estimated, $R \times C$ parameters (the size of the matrix) is greater than the number of distinct constraints that are applied, $R + C - 1$ constraints, we have an under-determined system. This issue becomes more problematic the more categories one has, due to the number of free parameters increasing. To illustrate, suppose a region is composed of 60% religious people and the other 40% atheist. We see that the votes of a policy have a 65% of support. Without more information, one cannot definitively say how the religious belief affected their decision. It is possible that religious population supported the policy almost unanimously. It is also possible that it was not as well liked among the religious population and only half of them gave support. Furthermore, it could even happen that the religious beliefs of the people had no effect whatsoever, so both groups supported it at similar rates. Each of these scenarios would result in the same overall numbers. This ambiguity is the main issue in ecological inference, as choosing one of those possibilities over the others without proper justification leads to an ecological fallacy (the mistaken assumption that relationships observed for groups necessarily hold for individuals within those groups), an issue first recognised by Robinson[1].

Another alternative to this issue is, instead of performing inference on the aggregated data, to measure the real individual behaviour. This completely circumvents the issue of a possible ecological fallacy, making the use of ecological inference redundant, but sadly it is not always possible to do. In many areas; like social sciences, public health, and politics the only available information is the marginal data. In those areas, where it is common to be working with sensitive data, only the aggregate data is made public (if at all) to ensure the privacy of the population. Additionally, it can also happen that, because no individual level survey was performed at the time, it's no longer possible to collect samples of individual behaviour.

1.1 History of the Ecological Inference Problem

The two first solutions, proposed in 1953, were by Duncan and Davis with the bounds method[2] and by Goodman and Leo with an ecological regression model[3][4]. In the bounds method the range of possible values is computed and that is the final answer. On the one hand, this approach guarantees no error (as your solution is every valid solution). On the other hand, the result found by this method can offer too much uncertainty in the intervals, making it impractical for multiple uses. Alternatively, the solution proposed in the ecological regression model is to get the aggregated data of multiple regions and tie their results together. To do this, some assumptions about the data distribution need to be made and in their 1953 paper, Goodman and Leo proposed that the parameters to estimate were the same across all regions[3]. In 1997, King showed that Goodman and Leo's assumptions, although sufficient for an unbiased result, were too strict for most applications, without a clear way to verify they hold[5]. Thus, when he published his paper, he changed it to the following:

1. The values of the transition matrix follow a truncated normal distribution (values outside the $[0, 1]$ interval are rejected), although later relaxed in his 1999 paper to any unimodal probability distribution (a probability distribution function with a single local maxima or mode) for Monte Carlo simulations[6].
2. The aggregates from the outcome grouping of two different regions are independent if the value of the aggregates of the initial grouping of one of those regions is known. This is also known as conditional independence[7] ($P(A|B \cap X) = P(A|X)$).
3. The mean of the values of the transition matrix are independent of the totals of the first grouping, later changed to the values being independent (not only the mean) in 1999[6].

Other assumptions have also been proposed, like Thomsen did in 1987[8]. In the paper, Thomsen proposed two things. The first, that the reasons for someone making a specific choice (be it the first or second grouping) could be expressed with latent variables. The second, that those latent variables follow a normal distribution with constant variance across all levels of aggregation. Although King's proposed assumptions are not the only assumptions currently used, as the existence of the `ecolRxC` R package[9] demonstrates, it is the most common one.

Once chosen the assumption of the ecological regression, the new restrictions from the assumption transform the problem into an over-determined system, with the best solution being the one with the highest likelihood or lowest error. This error function is not unique across all the models as it can vary due to the assumptions to tie the results together or the likelihood of the data for the given model. Wakefield[10] shows a compilation of multiple ways to model the distributions for the case when the transition matrix is 2×2 .

Unfortunately, all three of King's assumptions could always be broken. Examples that violate each one of the assumptions can be seen below:

1. In a situation where a policy is being voted, and the policy has a different impact across regions (e.g., a social or economic policy), it is reasonable to expect that the level of support for the policy in each region will be positively correlated with how beneficial or impactful the policy is perceived to be in that region. For instance, if we are studying the relationship

between race and policy support, and the policy's effects are bimodal across regions (e.g., it helps certain communities while disadvantaging others), the distribution of the cells of the transition matrix may not follow a unimodal distribution, as King's assumptions would expect.

2. In a presidential election, imagine that one of the candidates has greater support among certain religious groups. If we are comparing two regions that have similar age group distributions but differ significantly in the religious groups, we will observe different levels of support for that candidate in each region, even if the other factors are the same. This implies that the level of support for the different parties will not be conditionally independent given the distribution of age. Thus, King's second assumption, that the final aggregates are conditionally independent on the initial aggregates does not hold.
3. Consider a supermarket that offers a variety of products, such as fizzy drinks, along with other types of beverages. It is likely that the variety of fizzy drink options available in the supermarket will be positively correlated with the number of customers who buy them. For example, in a neighbourhood with a high proportion of students, who may have a stronger preference for sugary or caffeinated drinks, the supermarket is more likely to offer a wider selection of fizzy drinks. In contrast, in a region with a more elderly population or simply a population that prefers healthier beverages, the supermarket may offer a more limited range of fizzy drinks. Then the absence of one of the options or the over-representation of the other could influence the customers choice towards the more common product over what they would normally choose. This correlation between customer choice and product variety makes that areas with lower fizzy drink consumption have a lower preference for them and vice versa. This mean that the counts of students will be correlated to the preference of fizzy drinks on students and thus break the third and last assumption of King that said they were independent.

For scenarios where one of these assumptions is broken like could be in the above examples, three new methods have been proposed that relax or circumvent some of the possible restrictions that come with King's assumptions. These methods are the following:

- **Bayesian Ecological Inference Model with Covariates:** This method, introduced by King, Rosen, and Tanner, focuses on ecological inference for the 2×2 case, incorporating covariates into the model[6]. The inclusion of covariates allows for more nuanced estimates by accounting for additional factors that may affect the outcomes, improving the model's precision. In this context, the method helps to address situations where variables, that are linearly correlated with the transition matrix, would otherwise break some of the assumptions. Later, King, Rosen, Tanner, and Jiang extended this approach to arbitrarily big $R \times C$ tables[11].
- **Bayesian Ecological Inference Model with Spatial Effects:** Haneuse and Wakefield's approach takes into account spatial correlations by incorporating geographical proximity between regions (whether they are neighbours or not) in the 2×2 model[12]. In their ecological inference method, they use this information to model spatially structured random

effects with the help of a conditional autoregressive normal distribution[13] (CAR-normal) where each value depends on its neighbours. This spatial structure is important because neighbouring regions often exhibit correlated behaviours due to shared environmental, demographic, or economic factors. By incorporating this effects into the model, Haneuse and Wakefield’s method relaxes the assumption of independence between regions, acknowledging that spatial autocorrelation, correlations between nearby regions due to geographical closeness, can influence the inference. This approach provides a more realistic model for ecological data, especially in cases where the physical or social proximity of regions plays a significant role in shaping outcomes.

- **Bayesian Ecological Inference Method with Clustering:** Puig and Ginebra method, that works for arbitrary $R \times C$ tables, defines a model that allows the data to be grouped into clusters[14][15]. These clusters are determined by the model itself, meaning that the inference process dynamically adapts to the data structure. Each cluster is treated as having its own distribution for the parameters of the transition matrix. This approach relaxes the assumption of a convex unimodal distribution to a multimodal distribution with K modes, where K is the number of clusters identified by the model. This flexibility enables the model to account for heterogeneous patterns within the data and can lead to more accurate and insightful ecological inferences when the data exhibits clearly defined different patterns in different subgroups or regions.

1.2 Objectives

The goal of this paper is twofold: first, to propose a refined ecological inference model that extends the multinomial-Dirichlet framework proposed by King, Rosen, Tanner and Jiang in their 2001 paper[11]; and second, to evaluate the performance of this new model against existing approaches using simulated and real-world data. Building on the foundational work by King, Rosen, Tanner, and Jiang, we introduce an alternative formulation of the covariate model that retains the key strengths of the original, namely, its flexibility and more relaxed assumptions, while addressing some of its structural limitations.

Beyond proposing this revised model, an additional objective of the paper is to evaluate one common modelling assumption: the inclusion (or exclusion) of random effects. While many existing models include such effects by default, few studies have explicitly examined the consequences of this choice. We address this gap by implementing two versions of the model, one with covariates only, and one that augments the covariate model with random effect, and comparing their performance. This allows us to assess whether excluding random effects leads to meaningful gains in interpretability, computational efficiency, or convergence, and whether such simplifications risk sacrificing robustness or accuracy when unobserved heterogeneity is present.

To assess the practical utility of our proposed model, we benchmark its performance against a set of well-established ecological inference methods, both Bayesian and numerical ones. This includes the original model with covariates as described by King, Rosen, Tanner, and Jiang. We examine the method’s behaviour under a variety of scenarios, including different levels of noise (heterogeneity), and covariate relevance. These comparisons should also provide insight into

when the added complexity of covariates or latent structure is justified, and when a simpler model may suffice.

Overall, this paper contributes to the ecological inference literature by offering a theoretically sound, empirically validated, and practically interpretable model for incorporating covariates into transition matrix estimation, along with a careful examination of its strengths and limitations relative to existing alternatives.

2 Notation

To facilitate clarity and comparison across the various ecological inference methods discussed in this paper, we introduce a unified notation system. Ecological inference literature often varies in how key variables and structures are named and defined, which can obscure similarities and differences between approaches. By establishing a consistent set of symbols and definitions for this paper, we aim to streamline the exposition and enable direct methodological comparisons.

This section consolidates the notation used throughout the paper, covering both the foundational components shared by traditional methods and the extensions introduced in each of the models covered. We also specify the visual cues used to distinguish the origin, structure, and dimensionality, where applicable. This unified framework ensures that all subsequent explanations are grounded in a common language.

2.1 Data Origin

When defining a model, the meaning of parameters will be the following:

- Uppercase letters (X, Y): will be used for values related to the data.
- Lowercase letters (x, p): will be used for model hyperparameters and variables defined in the model.
- Greek letters (χ, δ): will be used for the values estimated by the model.

2.2 Data Shape

When defining a model or in formulas, the shape of the parameters will be the following:

- Normal letters (X, R): will be used for scalar values.
- Arrow over letters ($\vec{X}, \vec{N}_{1:T}$): will be used for vectors.
- Bold letters ($\mathbf{X}, \beta_{1,1:R,1:C}$): will be used for matrix or higher dimension values.

When using subindex, the shape will be according to the result. A single value will be to select only that element while two values separated by a colon ($a : b$) will signify a range. So $\vec{X}_{i,1:J}$ would mean the vector formed by the first element along the first dimension and all elements from 1 to J included along the second dimension (i.e. the vector formed by the first J elements of the i^{th} row).

2.3 Naming

If multiple parameters would share the same name, a super-index parenthesis will be used to differentiate between them ($X^{(a)}$, $Y^{(1)}$). The possible super-indexes are:

- By the first grouping or by rows: $\square^{(1)}$
- By the second grouping or by columns: $\square^{(2)}$
- At the individual level: $\square^{(i)}$
- At the global level: $\square^{(G)}$

For the specific names, the definitions are as follows:

Constant Scalars

- T : Number of regions. Alternatively number of tables.
- R : Number of elements in the first or initial grouping. Alternatively, the number of rows of the tables.
- C : Number of elements in the second or outcome grouping. Alternatively, the number of columns of the tables.
- P : Number of covariates present.
- k (only in the cluster model): Number of clusters to use.

Constant Non-Scalars

- $G^{(1)}$, $G^{(2)}$: The set of options or categories for the first (or initial) and second (or outcome) groupings.
- $\mathbf{Y}_{1:T, 1:R, 1:C}$: Counts of the elements from the crossed grouping. Alternatively, the filled tables. This value is unknown.
- $\mathbf{Y}_{1:T, 1:R}^{(1)}$ and $\mathbf{Y}_{1:T, 1:C}^{(2)}$: Counts of the elements from either the first or second grouping respectively. Alternatively, the sum over the rows or columns of the tables respectively.
- $\mathbf{p}_{1:T, 1:R}^{(1)}$ and $\mathbf{p}_{1:T, 1:C}^{(2)}$: The values of $\mathbf{Y}_{1:T, 1:R}^{(1)}$ and $\mathbf{Y}_{1:T, 1:C}^{(2)}$ row normalised respectively to sum 1.
- $\vec{N}_{1:T}$: Population size of a region. Alternatively the sum of the values of $\mathbf{Y}$, $\vec{Y}^{(1)}$ or $\vec{Y}^{(2)}$. If the sums can differ, $\vec{N}_{1:T}^{(1)}$ and $\vec{N}_{1:T}^{(2)}$ respectively.
- $\mathbf{X}_{1:T, 1:P}$: Covariates by region.
- $\mathbf{ADJ}_{1:T, 1:T}$ (only in the spatial model): The adjacency matrix is a matrix with values either one or zero. A one at (t_1, t_2) indicates that t_1 and t_2 are neighbours and a zero indicates that they aren't.

Model Estimates and Corresponding Priors

- $\beta_{1:T, 1:R, 1:C}$ and $b_{1:k, 1:R, 1:C}$ (only in cluster model): Coefficients of the transition matrix ($\beta_{i,j} \approx P(G^{(2)} = j \mid G^{(1)} = i)$) and the parameter for its prior b .
- $\theta_{1:k, 1:R}^{(1)}$ (only in cluster model), $\theta_{1:T, 1:C}^{(2)}$, $a_{1:k, 1:R}$ (only in cluster model): The probabilities for an element to belong to each group in the first and second grouping respectively ($\theta_r^{(1)} \approx P(G^{(1)} = r)$ and $\theta_c^{(2)} \approx P(G^{(2)} = c)$) and the parameter a for the prior of $\theta^{(1)}$.
- $\delta_{1:R, 1:C}$ and $d_{1:R, 1:C}$: Mean or intercept of the transition matrix in a transformed space ($f(\mathbb{E}[\beta]) = \delta$) and the parameter for its prior, d . As most instances in this paper will be the logit link function (except for the latent model which uses a probit), it will usually be $\delta = \text{logit}(\mathbb{E}[\beta])$.
- $\gamma_{1:R, 1:C}$ and $g_{1:R, 1:C}$: Coefficients of a covariate in a transformed space ($f(\mathbb{E}[\beta]) = \delta + \gamma Z$) and the parameter for its prior, g . As most instances in this paper will be the logit link function (except for the latent model which uses a probit), it will usually be $\delta + \gamma = \text{logit}(\mathbb{E}[\beta])$.
- $\nu_{1:T, 1:2}$ (only in the spatial model): Random effects of the model in the logit transformed space.
- $\vec{v}_{1:T}$ (only in the spatial model): Structured effects of the model in the logit transformed space.
- $\vec{\zeta}_{1:k}$ (only in the cluster model): Cluster number of the table.
- $\vec{\omega}_{1:k}$ and $\vec{o}_{1:k}$ (both only in cluster model): Probabilities of the regions belonging to a cluster and the parameter for its prior $\vec{o}$.
- σ and τ : The standard deviation and the precision (inverse of the variance) respectively.
- λ : A multiplier to scale precision.

Functions

- $\text{logit}(x)$ and $\text{multinomial logit}(\vec{x})$: The logit link function and the multidimensional version the multinomial logit link function:

$$\text{logit}(x) = \log\left(\frac{x}{1-x}\right)$$

$$\text{multinomial logit}(\vec{x})_i = \log(x_i) - \frac{1}{d} \sum_{j=0}^d \log(x_j)$$

These functions allows to transform probabilities to the reals ($\mathbb{R}$).

- $\text{Beta}(a, b)$, $\text{Binomial}(p, n)$, $\text{Categorical}(\vec{p})$, $\text{Dirichlet}(\vec{a})$, $\text{Exponential}(\lambda)$, $\text{Gamma}(\mu, r)$, $\text{Logistic}(\mu, r)$, $\text{Multinomial}(\vec{p}, n)$, $\text{Normal}(\mu, \tau)$, $\text{CAR-Normal}(\mathbf{ADJ}, \tau)$ and $\text{Flat}()$ (equivalent to a $\text{Uniform}(-\infty, \infty)$): Probability distributions (with the last two being improper).

		Voting behaviour			
Party:		A	B	C	
Age group	18 to 29	10	30	70	$Y_1^{(1)} = 110$
	30 to 49	55	20	50	$Y_2^{(1)} = 125$
	50 to 79	80	30	5	$Y_3^{(1)} = 115$
	80 and up	5	60	10	$Y_4^{(1)} = 75$
		$Y_1^{(2)} = 150$	$Y_2^{(2)} = 140$	$Y_3^{(2)} = 135$	$N = 425$

Table 1: Example table of age against party support

2.4 Example

Given the example in table 1, the values of the ecological inference problem would be the following:

- The categories for each grouping are the following:

$$G^{(1)} = \{18 \text{ to } 29, 30 \text{ to } 49, 50 \text{ to } 79, 80 \text{ and up}\}$$

$$G^{(2)} = \{\text{Party A, Party B, Party C}\}$$

- The size of the problem is $R = \|G^{(1)}\| = 4$ by $C = \|G^{(2)}\| = 3$.
- The unknown table and the corresponding transition matrix are these:

$$\mathbf{Y} = \begin{pmatrix} 10 & 30 & 70 \\ 55 & 20 & 50 \\ 80 & 30 & 5 \\ 5 & 60 & 10 \end{pmatrix} \Rightarrow \boldsymbol{\beta} = \begin{pmatrix} \frac{1}{11} & \frac{3}{11} & \frac{7}{11} \\ \frac{55}{125} & \frac{20}{125} & \frac{50}{125} \\ \frac{80}{115} & \frac{30}{115} & \frac{5}{115} \\ \frac{5}{75} & \frac{60}{75} & \frac{10}{75} \end{pmatrix}$$

- The total counts, also known as the marginals of the table, look like so:

$$\vec{Y}^{(1)} = (110, 125, 115, 75)^T$$

$$\vec{Y}^{(2)} = (150, 140, 135)^T$$

- The size of the population is $N = 425$.

3 Previous approaches

This section provides a comprehensive review of the key methodological developments in the field of ecological inference, focusing especially on approaches that have been proposed to ad-

dress the limitations inherent in King's original framework. While King's model has been foundational and widely influential, it relies on several assumptions that could be broken in certain applications, potentially limiting its accuracy or applicability. Consequently, a variety of alternative methods have emerged, aiming either to relax these restrictive assumptions or to enhance the robustness and precision of ecological inference estimates. In addition to modifications of King's model, this section also explores innovative techniques and recent advances that have demonstrated comparable or even superior performance in empirical studies.

3.1 Bayesian Model with Covariates

This model was first presented in the paper by King, Rosen, and Tanner[6], where they proposed a binomial-beta Bayesian hierarchical model for the 2×2 case of ecological inference. The model enables estimation on individual-level behaviour from grouped data using Markov Chain Monte Carlo (MCMC) methods, with the key innovation being the incorporation of covariates.

What all this means is the following: First, a Bayesian model is a statistical model that describes some phenomenon and can be updated to better reflect reality when new data becomes available through the use of Bayes theorem. As a closed form is not always feasible or computationally useful to obtain, an approximation can be made with Monte Carlo Markov chain simulations, generating a sample from the posterior of the model. Then, hierarchical refers to the fact that, in the statistical model, the parameters of some of the distribution depend on others. These distributions can then be grouped into how many layers or stages are between them and the final result. For example, if we have $x \sim \text{Uniform}(0, \rho)$, $\rho \sim \text{Exponential}(\lambda)$ and $\lambda \sim \text{Gamma}(1, 1)$; then the uniform would be in stage 1, the exponential in stage 2 and the gamma in stage 3. Finally, the binomial-beta part is the main choice of distributions made for representing the ecological inference problem.

The step by step of the model is the following:

First Stage, Observed Aggregates: To start, the observed counts of the second grouping are modelled using a Binomial distribution, as it describes the aggregated outcomes of the choice of a population between two options, exactly the structure seen in aggregated voting data in the 2×2 case. Although the binomial distribution only provides the number of times an event with probability p happened (the positive class), for example votes in favour of a policy, the counts for the opposite case (the negative class), the votes against said policy, can be easily calculated with the difference:

$$\begin{aligned} Y_{t,1}^{(2)} &\sim \text{Binomial}(N_t, \theta_t^{(2)}) \\ Y_{t,2}^{(2)} &= N_t - Y_{t,1}^{(2)} \end{aligned}$$

Where N_t is the total number of observed outcomes, and $\theta_t^{(2)}$ is the expected probability of someone to belong to the positive class in region t . These probabilities are computed by applying the transition matrix to the known probabilities of the first grouping like so: $\theta_t^{(2)} = \vec{p}_{t,1:2}^{(1)} \cdot \vec{\beta}_{t,1:2,1}$, where $\vec{p}_{t,1:2}^{(1)}$ is the proportions of the population when divided by the first grouping.

Second Stage, Covariate-Driven Transitions: Rather than assuming fixed transition probabilities, they model the positive class of each row of the transition matrices ($\vec{\beta}_{t,r}$) with a beta distribution, the conjugate prior of the binomial distribution. To incorporate the effect of the covariate into the model, they define the parameters of the Beta(a, b) distribution as follows:

$$\begin{aligned} a &= \sigma \cdot e^{(\delta + \gamma x)} \\ b &= \sigma \end{aligned}$$

This parametrization allows the expected value of the beta distribution to be expressed as a linear model with a logit link function:

$$\begin{aligned} \mathbb{E}[\text{Beta}(a, b)] &= \frac{a}{a + b} = \frac{\sigma \cdot e^{(\delta + \gamma X)}}{\sigma \cdot e^{(\delta + \gamma X)} + \sigma} = \frac{e^{(\delta + \gamma X)}}{e^{(\delta + \gamma X)} + 1} = \text{logit}^{-1}(\delta + \gamma X) \Rightarrow \\ &\Rightarrow \text{logit}(\mathbb{E}[\text{Beta}(a, b)]) = \delta + \gamma X \end{aligned}$$

This specification allows transition behaviour to vary flexibly across regions, depending on the covariate X_t . Each group of the initial grouping r has a corresponding intercept δ_r and covariate coefficient γ_r , which determine the expected direction and strength of covariate influence on transition probabilities. The scale parameter σ controls the concentration of the Beta distribution and acts similar to an inverse dispersion parameter.

Third Stage, Priors: To complete the model, they specify prior distributions for the higher-level parameters as follows:

$$\begin{aligned} \lambda &\sim \text{Exponential}(l) \\ \delta_r &\sim \text{Flat}(), \quad \forall r \in \{1, 2\} \\ \gamma_r &\sim \text{Flat}(), \quad \forall r \in \{1, 2\} \end{aligned}$$

Flat distributions as uninformative priors for δ and γ because the parameters can take any possible value, and an exponential for λ as it should only take positive values.

Extension

Although in the original paper only a proposal for the 2×2 case was made, two years later, the previous authors with the addition of Jiang, published a multinomial-Dirichlet hierarchical model[11]. This new model is a generalization of their binomial-beta method that allows for arbitrary $R \times C$ tables. This is achieved by swapping the binomial by a multinomial and the beta for a Dirichlet.

First Stage, Observed Aggregates: As mentioned, when upgrading from 2×2 to $R \times C$, the Binomial is swapped by a multinomial, a more general version that allows for multiple choices not just a binary one. With this new distribution, the probability of the positive class is changed to a probability vector, representing the likelihood of each choice. As the multinomial yields the counts for all classes, it can be expressed more compactly as follows:

$$\vec{Y}_{t,1:C}^{(2)} \sim \text{Multinomial}(N_t, \vec{\theta}_{t,1:C}^{(2)})$$

Where N_t is the total number of observed outcomes, and $\vec{\theta}_{t,1:C}^{(2)}$ is the expected probability vector for the choice of someone in region t . These probabilities are computed by applying the full transition matrix to the known probabilities of the first grouping like so: $\vec{\theta}_{t,1:C}^{(2)\top} = \vec{p}_{t,1:R}^{(1)\top} \times \beta_{t,1:R,1:C}$.

Second Stage, Covariate-Driven Transitions: Regarding the linear modelling of the covariate, one can first imagine the b parameter of the Beta(a, b) distribution to also use the linear model with coefficients fixed at zero. From there the following changes are applied to obtain the parameters for the Dirichlet distribution:

$$\begin{aligned} a &= \sigma \cdot e^{(\delta + \gamma x)} & \Rightarrow a_c &= \sigma \cdot e^{(\delta_c + \gamma_c x)} & \forall c \in \{1, \dots, C-1\} \\ b &= \sigma = \sigma \cdot e^{(0+0x)} & \Rightarrow a_C &= \sigma \cdot e^{(0+0x)} \end{aligned}$$

This similarly allows the expected value of the Dirichlet distribution to be rewritten as a linear model with a multinomial logit link function:

$$\begin{aligned} \mathbb{E}[\text{Dirichlet}(\vec{a})] &= \frac{\vec{a}}{\|\vec{a}\|_1} = \frac{\phi \cdot e^{(\vec{\delta} + \vec{\gamma}x)}}{\sum_{c=1}^C (\phi \cdot e^{(\delta_c + \gamma_c x)})} = \text{multinomial logit}^{-1}(\vec{\delta} + \vec{\gamma}x) \Rightarrow \\ &\Rightarrow \text{multinomial logit}(\mathbb{E}[\text{Dirichlet}(\vec{a})]) = \vec{\delta} + \vec{\gamma}x \end{aligned}$$

Third Stage, Priors: Regarding the priors, the only changes required are the addition of the fixed coefficients, resulting in this final distribution:

$$\begin{aligned} \lambda &\sim \text{Exponential}(l) \\ \delta_{r,c} &\sim \text{Flat}(), \quad \forall r \in \{1, \dots, R\}, c \in \{1, \dots, C-1\} \\ \gamma_{r,c} &\sim \text{Flat}(), \quad \forall r \in \{1, \dots, R\}, c \in \{1, \dots, C-1\} \\ \delta_{r,C} &= 0 \quad \forall r \in \{1, \dots, R\} \\ \gamma_{r,C} &= 0, \quad \forall r \in \{1, \dots, R\} \end{aligned}$$

Data transformation:

$$p_{t,r}^{(1)} = Y_{t,r}^{(1)} / N_t \quad \forall t \in \{1, \dots, T\}, r \in \{1, \dots, R\}$$

First stage:

$$Y_{t,1}^{(2)} \sim \text{Binomial}(N_t, \theta_t^{(2)}) \quad \forall t \in \{1, \dots, T\}$$

$$\hookrightarrow \vec{Y}_{t,1:C}^{(2)} \sim \text{Multinomial}(N_t, \vec{\theta}_{t,1:C}^{(2)})$$

$$\forall t \in \{1, \dots, T\}$$

$$\theta_t^{(2)} = \vec{p}_{t,1:R}^{(1)} \cdot \vec{\beta}_{t,1:R,1}$$

$$\forall t \in \{1, \dots, T\}$$

$$\hookrightarrow \theta_{t,c}^{(2)} = \vec{p}_{t,1:R}^{(1)} \cdot \vec{\beta}_{t,1:R,c}$$

$$\forall t \in \{1, \dots, T\}, c \in \{1, \dots, C\}$$

Second stage:

$$\beta_{t,r,1} \sim \text{Beta}(\sigma \cdot e^{(\delta_r + X_t \cdot \gamma_r)}, \sigma)$$

$$\hookrightarrow \vec{\beta}_{t,r,1:C} \sim \text{Dirichlet}(\sigma \cdot e^{(\vec{\delta}_{r,1:C} + X_t \cdot \vec{\gamma}_{r,1:C})})$$

$$\forall t \in \{1, \dots, T\}, r \in \{1, \dots, R\}$$

Third stage:

$$\lambda \sim \text{Exponential}(l)$$

$$\delta_{r,c} \sim \text{Flat}()$$

$$\forall r \in \{1, \dots, R\}, c \in \{1, \dots, C-1\}$$

$$\delta_{r,C} = 0$$

$$\forall r \in \{1, \dots, R\}$$

$$\gamma_{r,c} \sim \text{Flat}()$$

$$\forall r \in \{1, \dots, R\}, c \in \{1, \dots, C-1\}$$

$$\gamma_{r,C} = 0$$

$$\forall r \in \{1, \dots, R\}$$

Model 1: Covariate model, defined in section 3.1. Beta-binomial-specific parts are shown in gray, followed by the corresponding changes in the multinomial-Dirichlet case.

Given all that, the full Bayesian model can be seen in model 1.

A notable issue with this model is the choice of priors for the δ and γ parameters, flat priors with the final value fixed to zero for identifiability. As mentioned by Wakefield[10], for the ecological inference problem, the choice of a prior even on the second or third stage still yields a large impact on the final prediction. This becomes relevant when one realizes that the uninformed prior of the parameters that compose the transition matrix were chosen without taking into account the transformations that allow for the linear model to be used. If the transformation is applied on the flat improper prior and then normalised, the following cumulative distribution is obtained:

$$P(\mathbb{E}[\text{Beta}(a, b)] \leq p) = P(\text{logit}^{-1}(\delta + \gamma x) \leq p) = P(\delta + \gamma x \leq \text{logit}(p)) = \begin{cases} 1 & \text{if } p = 1 \\ \frac{1}{2} & \text{if } p \in [0, 1) \end{cases}$$

This result highlights that the transition matrix will be biased toward extreme values (zero and one). This behaviour also surfaces when using the multinomial-Dirichlet model, giving a value of one to a group and zero to the rest. Additionally, for the $R \times C$ case, as the ratio of known values to parameters to fit decreases, the effect of the uninformative prior, of tending towards more radical values, will be more noticeable on the results.

On top of that, the model lacks invariance under reordering of the outcome categories. For example, fitting age against party votes ordered as A, B, C versus B, C, A can lead to slightly different results. To understand the origin of this asymmetry, it is instructive to first walk through the generative process defined by the model:

1. **Initialization:** Vectors of mean and covariate effects, $\vec{\delta}_{r,1:C}$ and $\vec{\gamma}_{r,1:C}$, are initialised with random values, subject to an identifying constraint that fixes the final element of each vector to zero (to ensure identifiability).
2. **Transformation:** These parameters are combined additively and then exponentiated element-wise, resulting in a positive-valued vector $\vec{\rho}_{t,r,1:C}$, with the last element fixed at 1:

$$\begin{aligned} \rho_{t,r,c} &= e^{\delta_{r,c} + X_{t,c} \cdot \gamma_{r,c}} \\ \rho_{t,r,C} &= e^{\delta_{r,C} + X_{t,C} \cdot \gamma_{r,C}} = e^{0 + X_{t,C} \cdot 0} = 1 \end{aligned}$$

3. **Scaling:** The vector $\vec{\rho}_{t,r,1:C}$ is then scaled by a random scalar λ , drawn from an exponential distribution, to model the amount of random effect.
4. **Dirichlet Sampling:** The scaled vector forms the parameter vector of a Dirichlet distribution, [which governs the allocation of proportions]. The expected value of this Dirichlet distribution is the normalised vector $\frac{\vec{\rho}_{t,r,1:C}}{\|\vec{\rho}_{t,r,1:C}\|_1}$, and the variance is inversely related to $\lambda \cdot \|\vec{\rho}_{t,r,1:C}\|_1$. Thus, larger values of $\lambda \cdot \|\vec{\rho}_{t,r,1:C}\|_1$ correspond to lower variance and hence greater homogeneity in the inferred cell entries.

This construction introduces a dependence on the norm of $\vec{\rho}_{t,r,1:C}$, which is affected by the initial ordering of the second grouping because the fixed values for δ and γ are determined by that order. For example, consider a case in which parties A and B each command five times the support of party C. Then:

- Under the ordering of A, B, C we would get:

$$\bar{\rho}_{t,r,1:C}^{(ABC)} = \begin{pmatrix} 5 & 5 & 1 \end{pmatrix} \Rightarrow \left\| \bar{\rho}_{t,r,1:C}^{(ABC)} \right\|_1 = 11$$

$$\text{Var}(\beta_{r,c}^{(ABC)}) = f(\lambda^{(ABC)} \cdot 11)$$

- Under the ordering of B, C, A we would get:

$$\bar{\rho}_{t,r,1:C}^{(BCA)} = \begin{pmatrix} 1 & \frac{1}{5} & 1 \end{pmatrix} \Rightarrow \left\| \bar{\rho}_{t,r,1:C}^{(BCA)} \right\|_1 = \frac{11}{5}$$

$$\text{Var}(\beta_{r,c}^{(BCA)}) = f(\lambda^{(BCA)} \cdot \frac{11}{5})$$

Since the order of the labels should be independent of the variability between regions, we would expect $\frac{\lambda^{(ABC)}}{11} = \frac{5 \cdot \lambda^{(BCA)}}{11}$ to hold. For these two cases to be equally likely, an interval around the prior scaled in the same way ($[\text{low} \cdot \lambda; \text{high} \cdot \lambda]$) should yield the same probability for both cases. If we evaluate both probabilities, we reach the following equality:

$$L \sim \text{Exponential}(l)$$

$$\begin{aligned} P(L \in [\text{low} \cdot \lambda'; \text{high} \cdot \lambda']) &= P(L \leq \text{high} \cdot \lambda') - P(L \leq \text{low} \cdot \lambda') = \\ &= (1 - e^{-l \cdot \text{high} \cdot \lambda'}) - (1 - e^{-l \cdot \text{low} \cdot \lambda'}) = e^{-l \cdot \text{low} \cdot \lambda'} - e^{-l \cdot \text{high} \cdot \lambda'} = a^{\lambda'} - b^{\lambda'} \\ P(L \in [\text{low} \cdot \lambda^{(ABC)}; \text{high} \cdot \lambda^{(ABC)}]) &= P(L \in [\text{low} \cdot \lambda^{(BCA)}; \text{high} \cdot \lambda^{(BCA)}]) \Rightarrow \\ \Rightarrow a^{\lambda^{(ABC)}} - b^{\lambda^{(ABC)}} &= a^{\lambda^{(BCA)}} - b^{\lambda^{(BCA)}} \end{aligned}$$

Unfortunately, the last equality doesn't hold true for a fixed $a, b \in \mathbb{R}^+, a > b$ while λ is allowed to be any value ($\forall \lambda^{(ABC)}, \lambda^{(BCA)} \in \mathbb{R}^+$). This discrepancy implies that the effective prior over heterogeneity is not invariant to relabelling, potentially influencing posterior inferences. As such, care should be taken when interpreting or comparing model fits under different orderings, and researchers may consider standardizing outcome orderings or conducting a sensitivity analysis to quantify how much impact this has on the final outcome.

To summarize, this method enables both relaxation of the homogeneity assumption and testing for associations between covariates and behavioural change. However, it is not invariant to label ordering, and the uninformative priors can introduce a bias toward extreme values.

3.2 Bayesian Model with Spatial Dependencies

Ecological inference is often used to estimate the relationship between two categorical characteristics from aggregated data over regions or locations. Given that geographically proximate regions tend to exhibit a more closely linked behaviour due to greater interaction among their populations, Haneuse and Wakefield proposed enhancing the ecological inference model by incorporating spatial dependence into the model using geographical information[12].

Using data from Louisiana on race against party registration as their motivating example in their paper, they propose a hierarchical Bayesian model in which the variability of the transition matrix is decomposed into a structured spatial component, v ; an unstructured random component, ν ; and a mean effect δ .

First Stage, Observed Aggregates: To link the observed data and the transition matrix, they opt for approximating the effects of a binomial with a truncated normal distribution. This decision allows the model to still express the probabilities of someone making a choice with a reduced computational cost (albeit not much) of the model. Even though a normal distribution can generate values outside the $[0, 1]$ interval, the truncation is added to reject those values. As the normal distribution only generates one value, they use it to represent the proportion of one of the groups $p_{t,2}^{(2)}$, the second category of the outcome aggregates, while the proportion of the other outcome, $p_{t,1}^{(2)}$ the first category of the outcome aggregates, is described as the difference against the total:

$$p_{t,1}^{(2)} = 1 - p_{t,2}^{(2)}$$

$$p_{t,2}^{(2)} \sim \text{Normal}(\theta_{t,2}^{(2)}, \tau_{\theta,t}) \in [0, 1]$$

Here, $\theta_{t,2}^{(2)}$ is the expected probability of someone choosing the negative class, and $\tau_{\theta,t}$ is the precision (inverse of the variance) that a binomial would exhibit in the range $\{0, \dots, N_t\}$, normalised to the $[0, 1]$ (by dividing by N_t). Both of these values are obtained by the weighted sum of either the mean or inverse precision of a rescaled binomial with the weights of the two classes in the first grouping. Thus, those values can be expressed as follows:

$$\theta_{t,2}^{(2)} = p_{t,1}^{(1)} \cdot \frac{N_t \cdot \beta_{t,1,2}}{N_t} + p_{t,2}^{(1)} \cdot \frac{N_t \cdot \beta_{t,2,2}}{N_t} = p_{t,1}^{(1)} \beta_{t,1,2} + p_{t,2}^{(1)} \beta_{t,2,2}$$

$$\tau_{\theta,t}^{-1} = \sigma_{\theta,t}^2 = p_{t,1}^{(1)} \cdot \frac{N_t \cdot \beta_{t,1,1} \cdot \beta_{t,1,2}}{N_t^2} + p_{t,2}^{(1)} \frac{N_t \cdot \beta_{t,2,1} \cdot \beta_{t,2,2}}{N_t^2} =$$

$$= \frac{p_{t,1}^{(1)} \cdot \beta_{t,1,1} \cdot \beta_{t,1,2} + p_{t,2}^{(1)} \cdot \beta_{t,2,1} \cdot \beta_{t,2,2}}{N_t}$$

Where $\beta_{t,r,c}$ represents the probability of an individual in group r producing outcome c in region t .

The transition probabilities $\beta_{t,1,2,2}$ are then parametrised via a logit link to allow unconstrained modelling of probabilities (similar to what was done for the covariate model). This space allows them to decompose the variation in the transition matrix into different causes: global effects, local random effects, and spatially structured random effects. Still, they don't decompose both categories into the three effects, so the respective expressions look like so:

$$\text{logit}(\beta_{t,1,2}) = \delta_1 + \nu_{t,1}$$

$$\text{logit}(\beta_{t,2,2}) = \delta_2 + \nu_{t,2} + v_{t,2}$$

Here, δ_r is the global effect or intercept, $\nu_{t,r}$ a region-specific random effect for group r , and $v_{t,2}$ a spatially-structured effect capturing correlation across neighbouring units. Notably, only group 2 includes a spatial term, reflecting an asymmetry or prior knowledge about which group's behaviour exhibits spatial autocorrelation and only the second column of the transition matrix is calculated. The first column of the transition matrix (with complementary probabilities) can be obtained by subtracting the other from one ($\beta_{t,r,1} = 1 - \beta_{t,r,2}$)

Second Stage, Effect Distributions: For the mean effect δ_r , they opt for a logistic distribution (with parameters 0 and 1), which corresponds to a uniform prior on the original probability scale due to the properties of the logit transformation. In regard to the random effects $\nu_{t,r}$, they are modelled as independent normals ($\nu_{t,r} \sim \text{Normal}(0, \tau_{\nu_r})$). This is enough to capture non-spatial heterogeneity in the transition matrix, but it is better reinforced by the structured effects. Meanwhile, the spatially structured random effects $\nu_{t,2}$, they model it using a conditional autoregressive normal distribution[13] (CAR-Normal), a type of Gaussian distribution where their value are conditioned on one another.

This distribution can be written as:

$$\begin{aligned} X_i | X_j = x_j, \forall j \neq i, &\sim \mathcal{N} \left(\frac{\sum_{j \neq i} \mathbf{w}_{i,j} x_j}{\sum_{j \neq i} \mathbf{w}_{i,j}}, \frac{\hat{\tau}}{\sum_{j \neq i} \mathbf{w}_{i,j}} \right), & \forall i \\ X_i | X_j = x_j, \forall j \neq i, &\sim \mathcal{N} \left(\frac{1}{|\mathbf{N}(i)|} \sum_{j \in \mathbf{N}(i)} x_j, \frac{\hat{\tau}}{|\mathbf{N}(i)|} \right), & \forall i \text{ if } \forall i, j : \mathbf{w}_{i,j} \in \{0, 1\} \end{aligned}$$

Where $\mathbf{w}_{i,j}$ are weights encoding spatial proximity between regions i and j are, $\mathbf{N}(i)$ is the set of neighbours of i (those regions for which $\mathbf{w}_{i,j} = 1$) and $\hat{\tau}$ is proportional to the precision by some factor dependent on $\mathbf{w}_{i,j}$.

The choice for bell shaped distributions in the logit space instead of flat ones (like was done in the covariate model) is to take into account the distortion introduced by the logit link function, as a flat probability distribution ($\text{Uniform}(0, 1)$) gets most of its mass around zero ($\text{Logistic}(0, 1)$).

Third Stage, Priors: To complete the model, the authors use weakly-informative or hierarchical priors:

$$\begin{aligned} \delta_r &\sim \text{Logistic}(0, 1) & \forall r \in \{1, 2\} \\ \tau_{\nu_r} &\sim \text{Gamma}(a_{\nu, r}, r_{\nu, r}) & \forall r \in \{1, 2\} \\ \tau_{\nu_2} &\sim \text{Gamma}(a_{\nu, 2}, r_{\nu, 2}) \end{aligned}$$

The logistic prior on δ_r ensures the intercept is centred on 0 in logit space with moderate spread. Gamma priors on precisions offer flexibility while allowing data-driven shrinkage.

To balance the contributions of unstructured and structured variation, the model samples the precision parameters of both components from Gamma distributions. And for the choice of the parameters of the priors of the model: $a_{\nu, 1}$, $r_{\nu, 1}$, $a_{\nu, 2}$, $r_{\nu, 2}$, $a_{\nu, 2}$ and $r_{\nu, 2}$; they propose the following steps:

1. **Empirical Variance Ratio (ρ_{var}):** Compute the ratio between the variance of the CAR-normal distribution and the variance of the resulting samples generated for the given adjacency matrix.
2. **Expert-Based Confidence Interval:** Establish a 90% confidence interval for the values in the transition matrix using expert knowledge.

3. **Logit Transformation:** Apply the logit link function to the confidence interval.
4. **Total Marginal Variance (TMV):** Assuming approximate normality ($\text{logit}(\beta) \sim \mathcal{N}(\mu, \tau)$), compute the total marginal variance (TMV) implied by the interval $[\mu - \theta, \mu + \theta]$ using:

$$\text{TMV} = \left(\frac{\theta}{1.96} \right)^2$$

5. **Unstructured random effects prior parameters:** Determine Gamma prior parameters such that:
 - The inverse of the 5th percentile is equal to the TMV.
 - The inverse of the 95th percentile is a small value (e.g. 0.05, $\approx \frac{\text{TMV}}{20}$).
6. **Structured random effects prior parameters:** Similarly, select Gamma parameters such that:
 - The inverse of the 5th percentile is equal to the TMV over the empirical variance ratio $\left(\frac{\text{TMV}}{\rho_{\text{Var}}} \right)$.
 - The inverse of the 95th percentile is a small value (e.g. 0.05, $\approx \frac{\text{TMV}}{20 \cdot \rho_{\text{Var}}}$).

As the paper focuses exclusively on the Louisiana dataset of race against party registration, the proposed model (model 2) is tailored specifically to the 2×2 case, without a general framework for extending to $R \times C$ tables. This narrow focus also influences the choice of prior parameters, which rely heavily on expert knowledge. Although the authors describe a principled method for setting these priors, it relies on domain-specific knowledge, and they do not offer clear guidance for uninformative or data-driven alternatives. In our implementation, due to a lack of expert input, opted to assume the prior of the coefficients of the transition matrix to be a $\text{Beta}(1, 1)$, an uninformative prior, that when transformed with the logit link function leads to a $\text{Logistic}(0, 1)$. This means that its variance, and by extension the TMV, has a value of $\frac{\pi^2}{3}$. Compared to the previously discussed method (Bayesian Model with Covariates), Haneuse and Wakefield adopt a more deliberate approach in selecting prior distributions, explicitly avoiding bias toward the boundaries of the parameter space.

3.3 Bayesian Model with Clusters

One way to break the homogeneity assumption in ecological inference models is to assign independent parameters to each transition matrix. However, this would introduce a large number of parameters, possibly reducing the precision of the estimates. Moreover, many areas may still share similar voting behaviours, leading to redundancy in parameter estimation. To address this, Puig and Ginebra propose a clustering approach that groups areas with similar behaviour, striking a balance between flexibility and parsimony while effectively capturing spatial heterogeneity [14]. Notably, this clustering method requires no additional covariates, geographic or otherwise.

Data transformation:

$$p_{t,r}^{(1)} = Y_{t,r}^{(1)} / N_t \quad \forall t \in \{1, \dots, T\}, r \in \{1, 2\}$$

$$p_{t,c}^{(2)} = Y_{t,c}^{(2)} / N_t \quad \forall t \in \{1, \dots, T\}, c \in \{1, 2\}$$

First stage:

$$p_{t,2}^{(2)} \sim \text{Normal}(\theta_{t,2}^{(2)}, \tau_{\theta,t}) \quad \forall t \in \{1, \dots, T\}$$

$$\theta_{t,2}^{(2)} = p_{t,1}^{(1)} \beta_{t,1,2} + p_{t,2}^{(1)} \beta_{t,2,2} \quad \forall t \in \{1, \dots, T\}$$

$$\tau_{\theta,t} = \frac{N_t}{p_{t,1}^{(1)} \beta_{t,1,1} \beta_{t,1,2} + p_{t,2}^{(1)} \beta_{t,2,1} \beta_{t,2,2}} \quad \forall t \in \{1, \dots, T\}$$

$$\text{logit}(\beta_{t,1,2}) = \delta_1 + \nu_{t,1}$$

$$\text{logit}(\beta_{t,2,2}) = \delta_2 + \nu_{t,2} + v_{t,2}$$

$$\beta_{t,r,1} = 1 - \beta_{t,r,2} \quad \forall t \in \{1, \dots, T\}, r \in \{1, 2\}$$

Second stage:

$$\nu_{t,r} \sim \text{Normal}(0, \tau_{\nu_r}) \quad \forall t \in \{1, \dots, T\}, r \in \{1, 2\}$$

$$\vec{v}_{1:T,2} \sim \text{CAR-Normal}(\text{ADJ}, \tau_{v_2})$$

Third stage:

$$\delta_r \sim \text{Logistic}(0, 1) \quad \forall r \in \{1, 2\}$$

$$\tau_{\nu_r} \sim \text{Gamma}(a_{\nu,r}, r_{\nu,r}) \quad \forall r \in \{1, 2\}$$

$$\tau_{v_2} \sim \text{Gamma}(a_{v,2}, r_{v,2})$$

Model 2: Spatial model, defined in section 3.2.

First Stage, Observed Aggregates: Just like in the more general version of the covariate model, a multinomial distribution is used to describe the observed counts of the second grouping. Additionally, unlike the two previous discussed models, it also expresses the counts of the first grouping with a multinomial:

$$\begin{aligned}\vec{Y}_{t,1:R}^{(1)} &\sim \text{Multinomial}(N_t^{(1)}, \vec{\theta}_{t,1:R}^{(1)}) \\ \vec{Y}_{t,1:C}^{(2)} &\sim \text{Multinomial}(N_t^{(2)}, \vec{\theta}_{t,1:C}^{(2)})\end{aligned}$$

Where $\vec{\theta}_{t,1:R}^{(1)}$ and $\vec{\theta}_{t,1:C}^{(2)}$ represent the expected population proportions for both groupings, respectively, and $N_t^{(1)}$ and $N_t^{(2)}$ the total counts for both groupings.

Additionally, borrowing still from the covariate model, the expected probabilities for the second grouping are expressed as the application of the transition matrix on the first grouping proportions:

$$\vec{\theta}_{t,1:C}^{(2)\top} = \vec{p}_{t,1:R}^{(1)\top} \times \beta_{\zeta_t, 1:R, 1:C}$$

The difference in expression is the fact that each region t belongs to a latent cluster ζ_t , and the behaviour of each region in that cluster is governed by a shared transition profile $\beta_{\zeta_t, 1:R, 1:C}$. This structure enables information-sharing among regions that exhibit similar group-to-outcome patterns.

Second Stage, Cluster-Independent Transitions: In regard to the transition matrix, each row ($\vec{\beta}_{\zeta_k, r, 1:C}$) are modelled with Dirichlet, the conjugate prior of the multinomial used for the data. For the expected probabilities of the first grouping, they are also modelled with a Dirichlet prior for the same reasons. This means both can be written like this:

$$\begin{aligned}\vec{\beta}_{\zeta_k, r, 1:C} &\sim \text{Dirichlet}(\vec{b}_{\zeta_k, r, 1:C}) \\ \vec{\theta}_{t,1:R}^{(1)} &\sim \text{Dirichlet}(\vec{a}_{\zeta_t, 1:R})\end{aligned}$$

In both cases, the parameters of the priors are dependent on the cluster they get assigned to, facilitating convergence to a specific solution.

Third Stage, Latent Clustering: Each region t is assigned to one of k latent clusters (an arbitrary number chosen before running the model) through a categorical distribution.

$$\zeta_t \sim \text{Categorical}(\vec{\omega})$$

Where $\vec{\omega}$ defines the prior mixture weights over the k clusters. The chosen cluster (ζ_t) gets refined over time by the MCMC algorithm, forming stable clusters that explain the observed voting patterns. The variables also allows the model to capture the heterogeneity across regions, as it is the one grouping them into clusters with shared behavioural patterns.

Data transformation:

$$p_{t,r}^{(1)} = Y_{t,r}^{(1)} / N_t^{(1)} \quad \forall t \in \{1, \dots, T\}, r \in \{1, \dots, R\}$$

First stage:

$$\vec{Y}_{t,1:R}^{(1)} \sim \text{Multinomial}(N_t^{(1)}, \vec{\theta}_{t,1:R}^{(1)}) \quad \forall t \in \{1, \dots, T\}$$

$$\vec{Y}_{t,1:C}^{(2)} \sim \text{Multinomial}(N_t^{(2)}, \vec{\theta}_{t,1:C}^{(2)}) \quad \forall t \in \{1, \dots, T\}$$

$$\theta_{t,c}^{(2)} = \bar{p}_{t,1:R}^{(1)} \cdot \vec{\beta}_{\zeta_t, 1:R, c} \quad \forall t \in \{1, \dots, T\}, c \in \{1, \dots, C\}$$

Second stage:

$$\vec{\beta}_{k', r, 1:C} \sim \text{Dirichlet}(\vec{b}_{k', r, 1:C}) \quad \forall r \in \{1, \dots, R\}, k' \in \{1, \dots, k\}$$

$$\vec{\theta}_{t,1:R}^{(1)} \sim \text{Dirichlet}(\vec{a}_{\zeta_t, 1:R}) \quad \forall t \in \{1, \dots, T\}$$

Third stage:

$$\zeta_t \sim \text{Categorical}(\vec{\omega}) \quad \forall t \in \{1, \dots, T\}$$

Fourth stage:

$$\vec{\omega} \sim \text{Dirichlet}(\vec{\sigma}) \quad \forall t \in \{1, \dots, T\}$$

Model 3: Cluster model, defined in section 3.3.

Fourth Stage, Priors: Finally, the prior over cluster weights is given by the prior of the categorical distribution, the Dirichlet distribution:

$$\vec{\omega} \sim \text{Dirichlet}(\vec{\sigma})$$

All these steps together end up looking as follows (model 3):

An important thing to note from this method, is that it shows that a model even without individual random effects still yields sensible answers, although the accuracy of the results is never tested. Many ecological inference approaches assume homogeneity and rely on random effects to capture the unexplained variability at the individual level. In contrast, clustering similar areas together and estimating vote-switching behaviour within each group could account for a substantial part of the heterogeneity that would otherwise require an individual-level random effects.

Another strength of this model is its ability to clearly reveal when the assumption of homo-

geneity is broken. By clustering areas based on voting patterns, this method allows one to observe the emergence of groups with distinct transition matrices. If the clustering leads to significantly different transition patterns across regions, it provides direct evidence that the assumption that the areas are exchangeable does not hold, reinforcing the need for an alternative approach.

One of the challenges with this approach stems from the Markov Chain Monte Carlo (MCMC) results. This is because, when running multiple chains, they may not always converge to the clusters in the same order, complicating direct comparisons across runs. Additional post-processing is often required to match clusters between chains. Furthermore, the MCMC chains may not even converge to the exact same set of clusters in all the chains when the data allows for multiple plausible partitions. To help with this problem, the priors for the transition matrix, the row aggregates and the probabilities of each cluster allow for distinct parameters for each cluster. This means that if a more specific cluster solution is desired, these parameters can then be used to steer towards it, helping convergence of all chains into the same solution with the same representation.

In addition, the method requires an initial search for the optimal number of clusters that balances capturing spatial heterogeneity with avoiding over-fitting. This selection process often involves model comparison techniques, such as analysing residual spatial correlation or using posterior predictive checks, to identify the smallest number of clusters that sufficiently explain the data. While this step adds computational overhead, it is a necessary trade-off to ensure that the model captures meaningful spatial variation in the data.

3.4 Model with Latent Structure

The latent structure approach, originally proposed by Thomsen[8] and extended by Park[16], provides a causal framework for ecological inference with a different set of assumption by modelling voter behaviour based on unobserved (latent) policy preferences. The model assumes each voter has a set of preferences, represented in a latent space $\vec{L}_t^{(i)}$, following a normal distribution, with each region having a different mean $\vec{L}_t$ and a constant variance σ . Their individual voting decision (denoted with (i) superscript) in each election is determined by the interaction between this latent position $\vec{L}_t^{(i)}$, the stance of the candidates on those latent variables, $\vec{\gamma}$, and their popularity, δ . For the 2×2 case, the individual-level voting probability for the first class can be written as:

$$\begin{aligned} P(Y_{t,1}^{(1),(i)} = 1) &= \Phi(\delta_1 + \vec{\gamma}_1 \cdot \vec{L}_t^{(i)}) \\ P(Y_{t,1}^{(2),(i)} = 1) &= \Phi(\delta_2 + \vec{\gamma}_2 \cdot \vec{L}_t^{(i)}) \end{aligned}$$

Where Φ is the probit link function and $Y^{(1),(i)}$ and $Y^{(2),(i)}$ are the counts from the first and second election by individual (i.e. a one if they chose the first option and zero if they chose a different one). By integrating over the distribution of $\vec{L}_t^{(i)}$ (a normal distribution with mean L_t and variance σ), the expected proportions for group-level voting behaviour can be obtained like so:

$$\begin{aligned}
\mathbb{E} \left[p_{t,1}^{(1)} \right] &= \int_{\mathbb{R}^d} \Phi(\delta_1 + \vec{\gamma}_1 \cdot \vec{L}_t^{(i)}) \phi(\vec{L}_t^{(i)} | \vec{L}_t, \boldsymbol{\sigma}) d\vec{L}_t^{(i)} = \Phi \left(\frac{\delta_1 + \vec{\gamma}_1 \cdot \vec{L}_t}{\sqrt{1 + \vec{\gamma}_1^T \times \boldsymbol{\sigma} \times \vec{\gamma}_1}} \right) = \Phi \left(\delta'_1 + \vec{\gamma}'_1 \cdot \vec{L}_t \right) \\
\mathbb{E} \left[p_{t,2}^{(1)} \right] &= 1 - \mathbb{E} \left[p_{t,1}^{(1)} \right] \\
\mathbb{E} \left[p_{t,1}^{(2)} \right] &= \dots = \Phi \left(\frac{\delta_2 + \vec{\gamma}_2 \cdot \vec{L}_t}{\sqrt{1 + \vec{\gamma}_2^T \times \boldsymbol{\sigma} \times \vec{\gamma}_2}} \right) = \Phi \left(\delta'_2 + \vec{\gamma}'_2 \cdot \vec{L}_t \right) \\
\mathbb{E} \left[p_{t,2}^{(2)} \right] &= 1 - \mathbb{E} \left[p_{t,1}^{(2)} \right]
\end{aligned}$$

Where $p^{(1)}$ and $p^{(2)}$ are the proportions of populations for the first and second grouping respectively. The resulting formulas are the same as the individual with the parameters rescaled by $\sqrt{1 + \vec{\gamma}_i^T \times \boldsymbol{\sigma} \times \vec{\gamma}_i}$. Additionally, thanks to the assumptions of normality from the model and local independence[17] from the latent variables, the proportions (or probabilities) are a linear combination of normal distributions, thus they can be expressed as a binomial distribution:

$$p_{t,r,c} = \Phi_2 \left(\delta'_1 + \vec{\gamma}'_1 \cdot \vec{L}_t, \delta'_2 + \vec{\gamma}'_2 \cdot \vec{L}_t, \text{Corr} \left(\vec{\gamma}'_1 \cdot \vec{L}_t^{(i)}, \vec{\gamma}'_2 \cdot \vec{L}_t^{(i)} \right) \right) = \Phi_2 \left(\Phi^{-1}(p_{t,r}^{(1)}), \Phi^{-1}(p_{t,c}^{(2)}), \sigma'_{r,c} \right)$$

Where $p_{t,r,c}$ is the proportion of people that made both decisions, r for the first vote and c for the second one ($p_{t,r,c} = \frac{Y_{t,r,c}}{N_t}$); $\sigma'_{r,c}$ is the tetrachoric correlation for the (r, c) pair, obtained from the estimated tetrachoric correlation across regions; and Φ_2 is the bivariate normal cumulative function.

To generalize the method to $R \times C$ tables, the problem is divided into binary vote-switching problem. A reference cell is chosen ($r^{(\text{ref})}, c^{(\text{ref})}$) and for each other possible (r, c) pair, the 2×2 case is applied. The resulting cell estimates are then adjusted to align with the marginals using one of two iterative algorithms:

- Thomsen's Correction Method[8]
- Iterative Proportional Fitting[16]

The `ecolRxC` implementation[9] (the one used in this paper for the comparison) extends the latent structure approach by addressing several of its practical limitations. It introduces bootstrap-based estimation to provide uncertainty quantification in the form of confidence intervals for each cell of the transition matrix. To eliminate the arbitrariness associated with choosing a reference row and column in $R \times C$ tables, the model performs averaging over the solutions of all row-column reference pair combinations, a procedure they call Averaged Voter Correlation Reconstruction (AVCR). Furthermore, the method incorporates adjustments for turnout and census changes between elections, allowing for estimation with different population total counts. While the extension to $R \times C$ tables fundamentally relies on reducing the inference problem to multiple binary estimations, which can result in some loss of structural nuance, it nonetheless achieves performance comparable to other state-of-the-art alternatives, including King's Bayesian covariate model and the linear programming `nsphom` algorithm. On the other hand, performing all pair-wise combinations can significantly slow it down when the size of the transition matrix starts to grow.

3.5 Linear Programming Model

To frame the estimation of voter transitions as a constrained optimization problem, there is the nslphom algorithm, an advancement over the previously proposed tslphom, which itself improves upon the lphom model[18]. In order to understand how nslphom works, it is easier to begin by explaining its predecessors.

3.5.1 The lphom Model

The lphom model estimates a global transition matrix ($\beta_{1:R,1:C}^{(G)}$) instead of the transition matrix of each region. This is achieved via a linear programming approach grounded in the assumptions of ecological inference and an assumption of homogeneity across regions.

Under this framework:

- The rows of the transition matrix $\beta^{(G)}$ represent probability distributions, that is, their values are between 0 and 1, and they sum to 1.
- The transition matrix gives an exact result for the data aggregated over all regions.

$$\sum_{t=1}^T \left(\vec{Y}^{(1)}_{t,1:R} \right)^T \times \beta_{1:R,1:C}^{(G)} = \sum_{t=1}^T \left(\vec{Y}^{(2)}_{t,1:C} \right)^T$$

- By the homogeneity assumption, the transition matrix should also work for the different regions with some level of error, with the best solution being the one that yields the lowest error.

$$\vec{\epsilon}_{t,1:C} = \vec{Y}^{(2)}_{t,1:C} - \vec{Y}^{(1)}_{t,1:R} \times \beta_{1:R,1:C}^{(G)}$$

This yields the following formulation shown in model 4.

While quick and effective at producing global estimates, this model presents two key limitations:

- **Lack of local inference:** The solution is valid only at the aggregate (entire region) level, providing no direct information about the voter transitions at the level of individual territorial units.
- **Extreme value bias:** Due to the nature of linear programming, the optimal solution is often located at an extreme of the feasible region (a vertex in the convex hull), which leads to frequent estimates of 0 or 1, problematic in the context of transition probabilities.

3.5.2 The tslphom Model

To mitigate these issues, the tslphom algorithm introduces a two-step refinement. After getting an initial estimation by the lphom model, it first performs a localised adjustment to approximate the global solution within each region, and then re-aggregates these local estimates to construct an improved global transition matrix.

In that first step, for each region, the transition matrix that is *closest* to the global one and meets both of the ecological inference restrictions (that the rows are probability distributions

Minimize:

$$Z^{(1)} = \sum_{t=1}^T \sum_{c=1}^C |\varepsilon_{t,c}|$$

Subject to:

$$\varepsilon_{t,c} = Y_{t,c}^{(2)} - \sum_{r=1}^R Y_{t,r}^{(1)} \cdot \beta_{r,c}^{(G)} \quad \forall t \in \{1, \dots, T\}, c \in \{1, \dots, C\}$$

$$\beta_{r,c}^{(G)} \geq 0 \quad \forall r \in \{1, \dots, R\}, c \in \{1, \dots, C\}$$

$$\sum_{c=1}^C \beta_{r,c}^{(G)} = 1 \quad \forall r \in \{1, \dots, R\}$$

$$\sum_{r=1}^R \left(\sum_{t=1}^T Y_{t,r}^{(1)} \right) \cdot \beta_{r,c}^{(G)} = \left(\sum_{t=1}^T Y_{t,c}^{(2)} \right) \quad \forall c \in \{1, \dots, C\}$$

Model 4: Formulation for lphom, defined in section 3.5.1.

and $\vec{Y}^{(1)}_{t,1:R} \times \beta_{1:R,1:C}^{(G)} = \vec{Y}^{(2)}_{t,1:C}$ is selected. The *closest* being the one with the lowest absolute difference in the predicted number of votes, i.e. the difference weighted by the size of the population in each region. As just doing that doesn't yield a unique solution, from among the optimal solutions the one with the lowest absolute difference (without weights) is chosen. These two searches produces two formulations, where the second (model 6) depends on the first (model 6).

Once local solutions for each region are obtained, they are aggregated back to produce an updated global matrix:

$$\beta_{r,c}^{(G)} = \frac{\sum_{t=1}^T Y_{t,r}^{(1)} \cdot \beta_{t,r,c}}{\sum_{t=1}^T Y_{t,r}^{(1)}}$$

Because this aggregation reflects weighted behaviour across regions, it substantially reduces the tendency toward extreme values unless such values are consistently optimal in all units.

3.5.3 The nsiphom Model

Given the improvement that tslphom shows with respect to lphom[18], nsiphom doubles down on the idea, and repeats those refinement steps repeatedly. As successive steps don't converge towards a specific solution, a stopping criterion is used, the heterogeneity index, defined as:

Minimize:

$$Z_t^{(2)} = \sum_{r=1}^R \sum_{c=1}^C |\varepsilon_{t,r,c}|$$

Subject to:

$$\varepsilon_{t,r,c} = Y_{t,r}^{(1)} \cdot (\beta_{t,r,c} - \beta_{r,c}^{(G)}) \quad \forall r \in \{1, \dots, R\}, c \in \{1, \dots, C\}$$

$$\beta_{t,r,c} \geq 0 \quad \forall r \in \{1, \dots, R\}, c \in \{1, \dots, C\}$$

$$\sum_{c=1}^C \beta_{t,r,c} = 1 \quad \forall r \in \{1, \dots, R\}$$

$$\sum_{r=1}^R Y_{t,r}^{(1)} \cdot \beta_{t,r,c} = Y_{t,c}^{(2)} \quad \forall c \in \{1, \dots, C\}$$

Model 5: Formulation for the first step in tslphom part 1, defined in section 3.5.2

Minimize:

$$Z_t^{(3)} = \sum_{r=1}^R \sum_{c=1}^C |\Delta_{t,r,c}|$$

Subject to:

$$Z_t^{(2)} = \sum_{r=1}^R \sum_{c=1}^C |\varepsilon_{t,r,c}|$$

$$\Delta_{t,r,c} = \beta_{t,r,c} - \beta_{r,c}^G \quad \forall r \in \{1, \dots, R\}, c \in \{1, \dots, C\}$$

$$\varepsilon_{t,r,c} = Y_{t,r}^{(1)} \cdot (\beta_{t,r,c} - \beta_{r,c}^{(G)}) \quad \forall r \in \{1, \dots, R\}, c \in \{1, \dots, C\}$$

$$\beta_{t,r,c} \geq 0 \quad \forall r \in \{1, \dots, R\}, c \in \{1, \dots, C\}$$

$$\sum_{c=1}^C \beta_{t,r,c} = 1 \quad \forall r \in \{1, \dots, R\}$$

$$\sum_{r=1}^R Y_{t,r}^{(1)} \cdot \beta_{t,r,c} = Y_{t,c}^{(2)} \quad \forall c \in \{1, \dots, C\}$$

Model 6: Formulation for the first step in tslphom part 2, defined in section 3.5.2

$$HETe = \frac{100}{2} \cdot \frac{\sum_{t=1}^T \sum_{r=1}^R \sum_{c=1}^C Y_{t,r}^{(1)} \cdot |\beta_{t,r,c} - \beta_{r,c}^{(G)}|}{\sum_{t=1}^T N_{t\vec{\gamma}'_2\vec{L}^{(i)}}$$

The two proposed ways to use it are:

- First Minimum Rule: Stop at the first iteration n for which $HETe_n > HETe_{n-1}$
- Fixed Iteration Rule: Run for a fixed number N of iterations and select the iteration n with the minimum $HETe$.

4 Proposed Solution

A common assumption in ecological inference is regional homogeneity, that is, the assumption that all regions exhibit similar transition patterns, regardless of contextual effects such as neighbouring regions or socioeconomic factors. While this simplifies model structure and estimation, it can be overly restrictive in some cases, potentially leading to biased results when meaningful spatial or contextual variation is present. One effective way to relax this constraint is to incorporate covariates into the model, as was done in the covariate model by King, Jiang, Rosen, and Tanner. This strategy introduces flexibility by allowing different areas to respond differently depending on observed external variables. Additionally, incorporating covariates enables the model to estimate meaningful correlations between specific covariates and elements of the transition matrix, offering interpretability and insight into possible drivers of variation. The original covariate model has already shown that such an approach produces sensible results, lending credibility to further developments based on its structure. For these reasons, we propose a new Bayesian multinomial-Dirichlet hierarchical model that incorporates covariates, using the framework proposed by King, Rosen, Tanner, and Jiang as a starting point.

First Stage, Observed Aggregates: Like in the base model, we opt to model the observed data with a multinomial distribution. The benefit of using a multinomial distribution is a natural fit, as it models the aggregated outcomes of individual-level categorical decisions over a finite set of options, exactly the structure seen in aggregated voting data. As the multinomial yields the counts for all classes, it can be expressed compactly as follows:

$$\vec{Y}_{t,1:C}^{(2)} \sim \text{Multinomial}(N_t, \vec{\theta}_{t,1:C}^{(2)})$$

Where N_t is the total number of observed outcomes, and $\vec{\theta}_{t,1:C}^{(2)}$ is the expected probability vector for the choice of someone in region t . These probabilities are computed by applying the full transition matrix to the known probabilities of the first grouping like so: $\vec{\theta}_{t,1:C}^{(2)\top} = \vec{p}_{t,1:R}^{(1)\top} \times \beta_{t,1:R,1:C}$.

Second Stage, Covariate-Driven Transitions: To model β and add random effects, the natural option is the Dirichlet distribution, the conjugate prior of the multinomial. The main thing to decide is how to connect the covariates with the transition probabilities. For this, we follow the same core idea as the covariate model, constructing a linear model of the covariates, but we diverge in how we transform this linear predictor into a valid probability vector. Instead of the asymmetric approach used in that model (where the last coefficient is fixed to zero), we adopt the logit link function, as used in the spatial model. The only change necessary is to switch to the more general version, the multinomial logit link function:

$$\text{multinomial logit}(\vec{x})_i = \log(x_i) - \frac{1}{d} \sum_{j=0}^d \log(x_j)$$

The multinomial logit generalizes the binary logit function to higher-dimensional categorical settings and has the advantage of being symmetric with respect to the ordering of categories. This symmetry is important to avoid artifacts introduced by arbitrary label choices, as discussed in section 3.1. Moreover, this transformation guarantees that the output is a valid probability vectors, making it a natural fit for modelling the rows of the transition matrix if we try to remove the random effects. All these can be expressed as just:

$$\vec{\beta}_{t,r,1:C} \sim \text{Diriclet} \left(\lambda \cdot \text{multinomial logit}^{-1} \left(\vec{\delta}_{r,1:C} + X_t \cdot \vec{\gamma}_{r,1:C} \right) \right)$$

Where γ and δ are the coefficient and intercept of the model respectively, and λ is a parameter to quantify the homogeneity of the model. This can be changed for a more complex linear model by swapping $X_t \cdot \vec{\gamma}_{r,1:C}$ with $\sum_{p=1}^P X_{t,p} \cdot \vec{\gamma}_{p,r,1:C}$ or whatever model is preferred.

Another important consideration introduced by the logit transformation is scale sensitivity. The logit of a probability is not linear (something twice as likely does not double the value in the logit space), and its behaviour depends heavily on the scale of the covariates. In particular, the variance and mean of each covariate can introduce substantial distortion into the model unless they are properly centred and scaled. We therefore standardize all covariates before including them in the model. This normalization helps to avoid the following issues:

1. **Non-zero Mean:** Covariates with a non-zero mean effectively shift the intercept (δ), pushing it into regions of the parameter space that may have low prior density. For instance, if a covariate with non-zero coefficient such as average income has a mean of 2000, the intercept must compensate by taking a value of similar magnitude but opposite sign. Since the uninformative prior on δ has zero mean, this forces posterior values toward extremes and can lead to biased (the prior prevents from fully compensating) or unstable inference.
2. **High Variance:** A covariate with large variance scales the linear predictor, exaggerating the effect of its associated coefficient. This leads the logit transformation to saturate near 0 or 1, biasing the resulting transition probabilities toward extremes even if the true effect is moderate.
3. **Low Variance:** Conversely, covariates with very small variance suppress the effect of their coefficients, potentially hiding meaningful correlations in the data. This makes the covariate effectively irrelevant, regardless of its true importance.

By centring and rescaling covariates to have zero mean and unit variance, we place them on a comparable scale, ensure stability of the model, and preserve interpretability of the model coefficients. Additionally, standardization prevents numerical overflows or zeros during MCMC sampling, which is particularly important when the model includes many covariates or is applied to large datasets. Still, it must be noted that in our proposal we do not currently account for interactions, correlations, or joint scaling effects among covariates. We also haven't tested that fixing the variance in this normalization to 1 will yield the best results, although it should still be an improvement. Future work may explore how combined variance affects the stability and interpretability of the linear predictor.

Third Stage, Priors: The use of a logit transformation also affects how we must specify priors for the linear model coefficients. As discussed earlier, a flat prior for the model coefficients in the transformed space can result in a prior over the probability space that is heavily biased toward extreme values (0 and 1). This is particularly problematic in sparse or unbalanced data settings, where the prior may dominate the posterior. To avoid this, we adopt a different strategy: rather than placing flat priors directly on the logit scale parameters, we place a flat prior in the original space of probability vectors using a Dirichlet distribution. We then transform this prior using the logit link, yielding a distribution over coefficients that corresponds to a uniform prior over the original probabilities. This approach avoids the need to artificially fix one parameter as it already yields a value with $C - 1$ degrees of freedom (what in the original version was achieved by fixing a coefficient to zero), where C is the number of outcome categories.

$$\begin{aligned}\vec{\delta}_{r,1:C} &\sim \text{multinomial logit}(\text{Dirichlet}(\vec{d}_{r,1:C})) \\ \vec{\gamma}_{p,r,1:C} &\sim \text{multinomial logit}(\text{Dirichlet}(\vec{g}_{p,r,1:C}))\end{aligned}$$

Alternative without random effects

In the clustering model, the omission of random effects yielded plausible estimates, suggesting that under certain conditions, explicit modelling of unobserved heterogeneity may not be strictly necessary. Removing random effects substantially reduces the number of parameters, which in turn lowers computational complexity, speeds up MCMC convergence, and simplifies interpretation by focusing inference on a smaller set of covariate-driven effects. However, without random effects, the model may become more sensitive to omitted variables, and potential errors could arise if the included covariates fail to capture all relevant sources of variation.

As the implications of this exclusion remain largely untested, we decided that modelling without random effects was an interesting choice worth exploring. Although in the clustering model some of this residual variation was absorbed via latent group assignments, it is unclear without tests whether covariates alone can serve this role effectively. To properly assess the trade-offs, we therefore implement and evaluate both versions of the model, with and without random effects. This allows us to empirically investigate the impact of this design decision and determine whether the simplifications achieved through exclusion compromise the quality or robustness of the estimates.

Data transformation:

$$p_{t,r}^{(1)} = Y_{t,r}^{(1)} / N_t^{(1)} \quad \forall t \in \{1, \dots, T\}, r \in \{1, \dots, R\}$$

$$z_{t,p} = \frac{X_{t,p} - \frac{1}{T} \sum_{t'=1}^T X_{t',p}}{\sqrt{\frac{\sum_{t'=1}^T (X_{t',p} - \frac{1}{T} \sum_{t''=1}^T X_{t'',p})^2}{T-1}}} \quad \forall t \in \{1, \dots, T\}, p \in \{1, \dots, P\}$$

First stage:

$$\vec{Y}_{t,1:C}^{(2)} \sim \text{Multinomial}(N_t^{(2)}, \vec{\theta}_{t,1:C}^{(2)}) \quad \forall t \in \{1, \dots, T\}$$

$$\theta_{t,c}^{(2)} = \bar{p}_{t,1:R}^{(1)} \cdot \vec{\beta}_{t,1:R,c} \quad \forall t \in \{1, \dots, T\}, c \in \{1, \dots, C\}$$

Second stage:

$$\vec{\beta}_{t,r,1:C} \sim \text{Dirichlet} \left(\lambda \cdot \text{multinomial logit}^{-1} \left(\vec{\delta}_{r,1:C} + \sum_{p=1}^P z_{t,p} \cdot \vec{\gamma}_{r,1:C}^{(p)} \right) \right) \quad \forall t \in \{1, \dots, T\}, r \in \{1, \dots, R\}$$

Third stage:

$$\vec{\delta}_{r,1:C} \sim \text{multinomial logit}(\text{Dirichlet}(\vec{d}_{r,1:C})) \quad \forall r \in \{1, \dots, R\}$$

$$\vec{\gamma}_{r,1:C}^{(p)} \sim \text{multinomial logit}(\text{Dirichlet}(\vec{g}_{r,1:C}^{(p)})) \quad \forall p \in \{1, \dots, P\}, r \in \{1, \dots, R\}$$

$$\lambda \sim \text{Exponential}(l)$$

Model 7: Our model with random effects, defined in section 4.

The change necessary for this new variation of the model is simple. Because the inverse of the logit transformation already return probabilities, we can plug those directly into the transition matrix like so:

$$\text{multinomial logit}(\vec{\beta}_{t,r,1:C}) = \vec{\delta}_{r,1:C} + X_t \cdot \vec{\gamma}_{r,1:C}$$

The resulting Bayesian model with random effects can be seen in model 7, while the model without is shown in model 8.

Data transformation:

$$p_{t,r}^{(1)} = Y_{t,r}^{(1)} / N_t^{(1)} \quad \forall t \in \{1, \dots, T\}, r \in \{1, \dots, R\}$$

$$z_{t,p} = \frac{X_{t,p} - \frac{1}{T} \sum_{t'=1}^T X_{t',p}}{\sqrt{\frac{\sum_{t'=1}^T (X_{t',p} - \frac{1}{T} \sum_{t''=1}^T X_{t'',p})^2}{T-1}}} \quad \forall t \in \{1, \dots, T\}, p \in \{1, \dots, P\}$$

First stage:

$$\vec{Y}_{t,1:C}^{(2)} \sim \text{Multinomial}(N_t^{(2)}, \vec{\theta}_{t,1:C}^{(2)}) \quad \forall t \in \{1, \dots, T\}$$

$$\theta_{t,c}^{(2)} = \vec{p}_{t,1:R}^{(1)} \cdot \vec{\beta}_{t,1:R,c} \quad \forall t \in \{1, \dots, T\}, c \in \{1, \dots, C\}$$

$$\text{multinomial logit}(\vec{\beta}_{t,r,1:C}) = \vec{\delta}_{r,1:C} + \sum_{p=1}^P z_{t,p} \cdot \vec{\gamma}_{r,1:C}^{(p)} \quad \forall t \in \{1, \dots, T\}, r \in \{1, \dots, R\}$$

Second stage:

$$\vec{\delta}_{r,1:C} \sim \text{multinomial logit}(\text{Dirichlet}(\vec{d}_{r,1:C})) \quad \forall r \in \{1, \dots, R\}$$

$$\vec{\gamma}_{r,1:C}^{(p)} \sim \text{multinomial logit}(\text{Dirichlet}(\vec{g}_{r,1:C}^{(p)})) \quad \forall p \in \{1, \dots, P\}, r \in \{1, \dots, R\}$$

Model 8: Our model without random effects, defined in section 4.

5 Comparison

To comprehensively evaluate the performance of each model, we conduct a series of quantitative comparisons across multiple dimensions. These comparisons are designed to assess both predictive accuracy and practical usability. To differentiate between the true data and the estimations of the model in the expressions for the metrics, we'll add a hat ($\hat{\square}$) to the model estimations, i.e. β for the true transition matrix, and $\hat{\beta}$ for the one estimated by the model. The list of things we measure is the following:

- **Individual-level Performance:** We evaluate the bias and deviation of each model's predictions at the individual level. This helps understand how close the predictions are to the actual values, and how consistently the model performs across different individuals. To calculate these two components, we calculate the mean and standard deviation for the error made in the estimation of $\beta_{t,r,c}$ over all regions. Additionally, to prevent regions with smaller population to have a greater effect (being off by one represents a greater change), we weight the calculation of the mean and deviation by the size of the group ($Y_{t,r}^{(1)}$) in that region, which can also be expressed as the unweighted calculation over all people, here expressed as the set $\mathcal{I}$. These two representations of the error are the following:

$$\text{Error}(\beta_{t,r,c}) = E_{t,r,c} = \hat{\beta}_{t,r,c} - \beta_{t,r,c}$$

$$\text{Bias}(E_{r,c}) = \frac{\sum_{i \in \mathcal{I}} E_{t_i,r,c}}{\|\mathcal{I}\|} = \frac{\sum_{t=1}^T Y_{t,r}^{(1)} \cdot E_{t,r,c}}{\sum_{t=1}^T Y_{t,r}^{(1)}}$$

$$\text{Deviation}(E_{r,c}) = \sqrt{\frac{\sum_{i \in \mathcal{I}} (E_{t_i,r,c} - \text{Bias}(E_{r,c}))^2}{\|\mathcal{I}\| - 1}} = \sqrt{\frac{\sum_{t=1}^T Y_{t,r}^{(1)} \cdot (E_{t,r,c} - \text{Bias}(E_{r,c}))^2}{\sum_{t=1}^T Y_{t,r}^{(1)} - 1}}$$

- **Bound Estimates:** As all the models generate uncertainty bounds $[\hat{\beta}_{t,r,c}^{(\text{low})}, \hat{\beta}_{t,r,c}^{(\text{high})}]$, we examine both the size of the confidence or credible intervals and their accuracy, i.e., how often the true values falls within the predicted bounds. For measuring the size of the bounds found by the model, we take the weighted average size of the bounds for each region, as regions with smaller population should have more ambiguity due to a difference of just one person representing a greater proportion. Meanwhile, for the accuracy we just calculate the frequency of times the true value ($\beta_{t,r,c}$) is within the confidence (or credible) intervals. For the accuracy we don't use the population size as the size of the bounds should already account for that. Then, the formulas for the two parameters look as follows:

$$\text{Size}(\beta_{r,c}) = \frac{\sum_{t=1}^T Y_{t,r}^{(1)} \cdot (\hat{\beta}_{t,r,c}^{(\text{high})} - \hat{\beta}_{t,r,c}^{(\text{low})})}{\sum_{t=1}^T Y_{t,r}^{(1)}}$$

$$\text{Accuracy}(\beta_{r,c}) = \frac{1}{T} \sum_{t=1}^T \begin{cases} 1 & \text{if } \beta_{t,r,c} \in [\hat{\beta}_{t,r,c}^{(\text{low})}, \hat{\beta}_{t,r,c}^{(\text{high})}] \\ 0 & \text{otherwise} \end{cases}$$

- **Region and Global Metrics:** We compute the Mean Squared Error (MSE) and Mean Absolute Error (MAE) of the predicted transition matrices, both at the regional or local level and globally. These metrics allow us to assess how well the models capture all patterns in the data and for which task they perform better. On top of using the formulas for the MAE and MSE, we normalize the results so we may compare results from different datasets, yielding the following expressions for the different errors:

$$\text{Local MAE} = \frac{\sum_{t=1}^T \sum_{r=1}^R \sum_{c=1}^C |\hat{Y}_{t,r,c} - Y_{t,r,c}|}{\sum_{t=1}^T N_t}$$

$$\text{Local MSE} = \frac{\sum_{t=1}^T \sum_{r=1}^R \sum_{c=1}^C (\hat{Y}_{t,r,c} - Y_{t,r,c})^2}{\sum_{t=1}^T N_t}$$

$$\text{Global MAE} = \frac{\sum_{r=1}^R \sum_{c=1}^C |\hat{Y}_{r,c}^{(G)} - Y_{r,c}^{(G)}|}{N^{(G)}}$$

$$\text{Global MSE} = \frac{\sum_{r=1}^R \sum_{c=1}^C (\hat{Y}_{r,c}^{(G)} - Y_{r,c}^{(G)})^2}{N^{(G)}}$$

- **Computational Efficiency:** We record the time taken to execute each model, providing insight into their practical feasibility for large-scale applications.
- **Model Inputs:** A record of the parameters passed to each model is maintained to ensure reproducibility and transparency in configuration.

Experimental Setup All tests were conducted on a SLIMBOOK EVO equipped with an AMD Ryzen 8845HS processor and 32GB of RAM, with the system set to the highest performance settings.

For the implementation, depending on the method, we did the following:

- **Bayesian Methods:** The chains of the Bayesian models were run in parallel using JAGS, except for the spatial method, which required WinBUGS for the CAR-normal distribution, and was therefore run in single-threaded mode.
- **Numerical Methods:** Numerical methods were executed using the original authors' R packages:

- Package `ecolRxC` (<https://cran.r-project.org/package=ecolRxC>) for the latent space model.
- Package `lphom` (<https://cran.r-project.org/package=lphom>) for the linear programming model.

Reproducibility All code used for these experiments is publicly available in the following GitHub repository: https://github.com/Littleote/TFM-Ecological_Inference/tree/TFM-Code
The repository includes:

- Full source code.
- The datasets used in the experiments.
- A README guide on how to reproduce the results and how the repository is structured.

5.1 Louisiana 2020 Presidential Election Results

For the first comparison of the different ecological inference models, we utilised the voter registration data from the state of Louisiana for the year 2020. The decision to use Louisiana as a test case was made taking the following into account:

- **Established Benchmark:** Data from Louisiana is frequently used in EI literature and applications, making it a standard and familiar benchmark for testing and validating models.
- **Availability of Ground Truth:** The state provides detailed voter registration data broken down by race and political party affiliation. This allows us to compare model estimates against known true values, which is essential for evaluating model performance against one another.
- **Size of the Transition Matrix:** The registration data lends itself to a 2×2 transition matrix with race either black or white and party Democrat or Republican. This is important because not all ecological inference models we described (specifically the [are](#) able to handle a more general $R \times C$ case (more than two races and parties). A 2×2 format ensures compatibility across all four methods and simplifies interpretation.

5.1.1 Data

The Louisiana registration data is a dataset containing both, the total number (also in percentage) of people per parish (county-equivalent) disaggregated by race party registration and parishes. The party registrations are divided into “Republicans”, “Democrats”, and “Other”, the aggregation of all other political affiliations. The races are broken down into three racial categories: “White”, “Black”, and “Other” (which, like before, is the aggregates of all other races). Registration data is reported at the parish level, with each parish providing the number of registrants by race and party affiliation. This granularity supports the use of ecological inference methods by providing sufficient variation across units.

This data is available because Louisiana is one of the few states that collects and publicly reports voter registration data disaggregated by both race and party. The practice stems from a

combination of administrative transparency and historical context, the Voting Rights Act policy that dates to 1965[19]. As a result, the state has maintained detailed records that are updated and published regularly by the Secretary of State, making it uniquely valuable for research requiring ground truth at the intersection of race and political affiliation. In contrast, most other states either do not collect racial data during registration or do not make it publicly accessible, often due to legal, political, or privacy concerns. This makes Louisiana an ideal case for ecological inference model validation, as it provides the rare opportunity to compare model estimates directly against known, disaggregated values.

Since the “Other” racial category represents a small fraction of the overall registration base, roughly one-tenth the size of the white population and one-fifth that of the black population, we excluded it from our analysis to ensure the construction of a more equilibrated 2x2 contingency table. For political affiliation, we focus on the two major parties in the United States: Democrats and Republicans. These are also the most prominent in the dataset, making them the most informative for our analysis.

To accommodate both ecological inference models with covariates, King’s and our own, we utilised a more recent year of Louisiana voter registration data (2020), ensuring that relevant covariate information was more easily available. In particular, we augmented the core dataset with parish-level median income and the difference in educational attainment between races.

Median income serves as a proxy for socioeconomic status, a factor that often correlates with political preference and may do so differently across racial groups. The inclusion of the racial education gap, defined as the difference in attainment between black and white residents, captures contextual inequalities that may shape political engagement and party affiliation. Together, these covariates should allow the models to better adjust to the underlying social and economic dynamics, particularly between otherwise homogeneous areas where racial composition alone would provide limited information.

Sources:

- Registration statistics for the state of Louisiana (web page: <https://www.sos.la.gov/ElectionsAndVoting/Pages/RegistrationStatisticsStatewide.aspx>, XLS: https://electionstatistics.sos.la.gov/Data/Post_Election_Statistics/statewide/2020_1103_sta.xls)
- Household income adjusted for inflation in Louisiana by parishes (table: [https://data.census.gov/table/ACSST5Y2020.S1901?g=040XX00US22\\$0500000](https://data.census.gov/table/ACSST5Y2020.S1901?g=040XX00US22$0500000))
- Education level in Louisiana by parishes and age or race (table: [https://data.census.gov/table/ACSST5Y2020.S1501?g=040XX00US22\\$0500000](https://data.census.gov/table/ACSST5Y2020.S1501?g=040XX00US22$0500000))
- US counties subdivision map (web page: <https://www.census.gov/cgi-bin/geo/shapefiles/index.php?year=2020&layergroup=County+Subdivisions>)

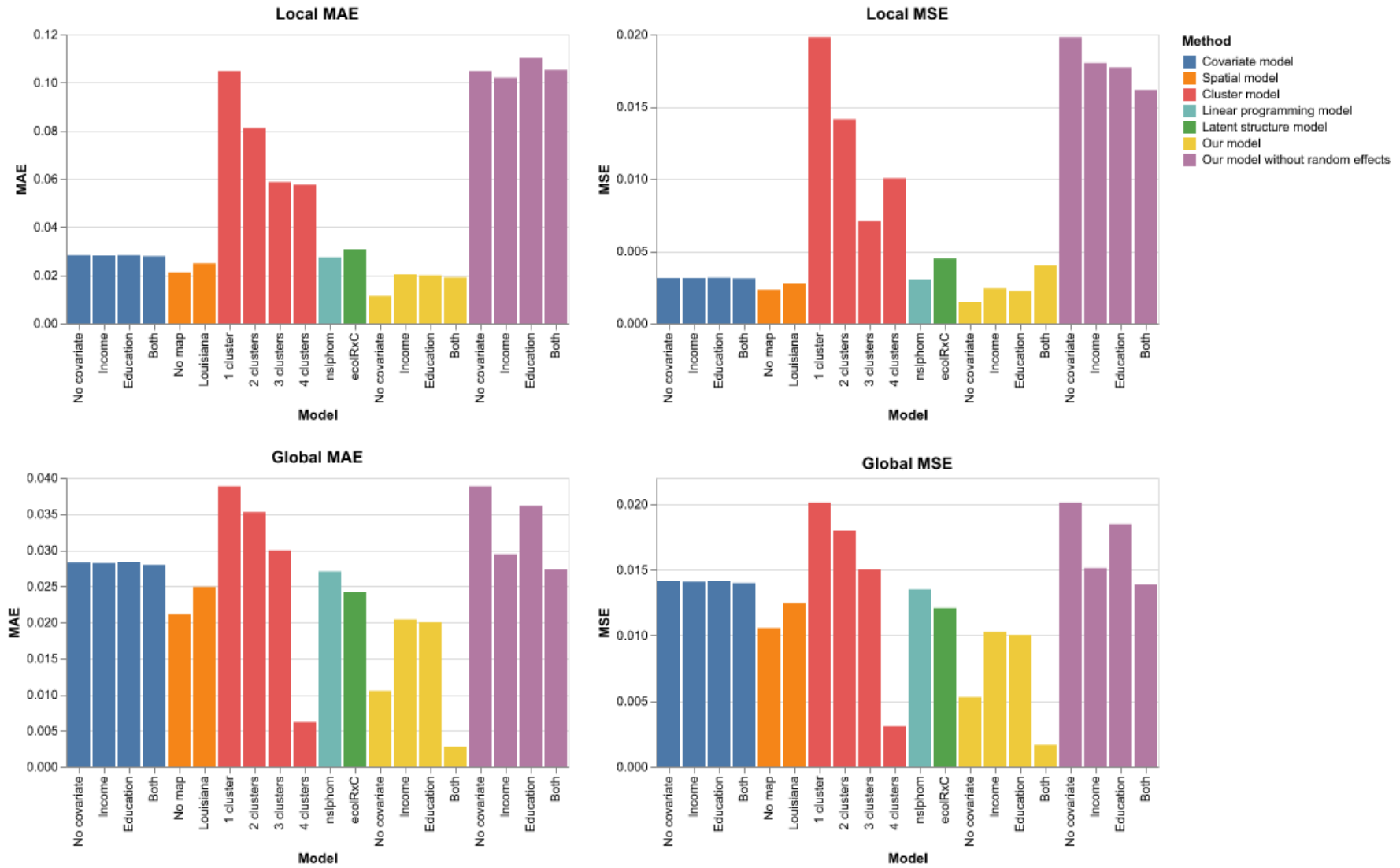


Figure 1: Error for the different models in the Louisiana data.

5.1.2 Results

The first thing that must be pointed out from the results is the fact that our model, with random effects and executed without any covariates, is the one yielding the best results of all. It has the lowest bias (table 2) and local error (top of figure 1). It also achieves the second-smallest bound size with a perfect match (table 3) and third-best global error (top of figure 1) Seeing this and the fact that adding more information worsens the results, leads to a clear conclusion. The priors or model are making the model tend towards the specific solution better in this specific case. Thus, this is an unfair comparison for this model and its results (other than time and the size of the bounds) should be taken with scepticism.

Besides that, the next thing that attracts attention is the lack of improvement (or any change at all) of King’s covariate model when adding covariates. For example, adding income or education covariates leads to marginal changes at best, but no meaningful gain with regard to bias or error. In some parameters, it even has convergence issues, specifically in the δ and γ chains for the black subgroups. Still, the posterior summaries of their combined result parameters, that is, of the target β transition matrices, appear stable for both races. This hints at a possible partial convergence due to the fact that the more extreme values of the black race make changes in the coefficient have an almost negligible effect. Apart from that, King’s covariate model achieves quite similar results with the two numerical methods. Still the time it takes to run, is three orders of magnitude bigger than the numerical methods and triple of our proposed mode.

Regarding the removal of random effects from the model, it shows a dramatically faster computation speed, around 10 times faster (e.g., 8.36 second against 87.89 seconds for the income case). However, this comes at the cost of a five-fold increase in the error of the local tables and barely matching the nsphom and covariate models for the global transition matrix error. On top of the more unreliable local estimates, the credible intervals obtained are too smalls to be of help. Both, the cluster model and our model without random effects, gave credible interval that never even covered 10% of the true values (table 3). All in all, these simpler models result unsuitable for inference at the level of regions despite their speed. Still, they are a useful tool for a task with a more global scope, about the decomposition into clusters the model makes (be it the cluster assignment $\vec{\zeta}$ or the transition matrix $\beta_{k', 1:R, 1:C}()$), or estimating the correlation matrix γ . For those scenarios it is helpful thanks to its shorter execution time and comparable error.

Finally, a demonstration of the predicted results for the best performing models of each method on Acadia (the first region alphabetically) can be seen on table 4

5.2 Artificial Data Based on New Zealand Elections

Ecological inference poses a unique validation challenge due to the general unavailability of datasets containing both aggregate data and their corresponding individual-level ground truth. As a result, many studies rely on a limited set of benchmark datasets, most notably the Louisiana parish-level data disaggregated by race, or resort to evaluating models on real-world data without known true values. However, for the Louisiana data, our model without covariates outperformed not only its covariate-augmented counterpart but also all other state-of-the-art ecological inference models. This unexpected result raises concerns that the strong performance may be

Method	Model	Bias($\beta_{1,1}$)	Bias($\beta_{2,1}$)	Execution time
Covariate model	No covariate	-0.010 ± 0.011	0.022 ± 0.015	161.92s
	Income	-0.010 ± 0.011	0.022 ± 0.015	96.30s
	Education	-0.010 ± 0.011	0.022 ± 0.015	138.41s
	Both	-0.010 ± 0.011	0.022 ± 0.015	188.08s
Spatial model	No map	-0.007 ± 0.008	0.017 ± 0.015	172.75s
	Louisiana	-0.009 ± 0.009	0.020 ± 0.015	239.75s
Cluster model	1 cluster	-0.018 ± 0.202	0.022 ± 0.015	13.83s
	2 clusters	-0.015 ± 0.132	0.022 ± 0.015	29.10s
	3 clusters	-0.012 ± 0.076	0.022 ± 0.016	38.55s
	4 clusters	0.002 ± 0.085	-0.005 ± 0.080	47.64s
Latent structure model	ecolRxC	0.009 ± 0.027	-0.019 ± 0.045	0.03s
Linear programming model	nsIphom	-0.010 ± 0.011	0.021 ± 0.016	0.75s
Our model	No covariate	-0.003 ± 0.008	0.009 ± 0.013	71.03s
	Income	-0.007 ± 0.010	0.018 ± 0.015	87.89s
	Education	-0.006 ± 0.009	0.018 ± 0.015	86.39s
	Both	-0.000 ± 0.022	0.004 ± 0.039	105.36s
Our model without random effects	No covariate	-0.018 ± 0.202	0.022 ± 0.015	5.59s
	Income	-0.013 ± 0.192	0.018 ± 0.038	8.36s
	Education	-0.016 ± 0.190	0.022 ± 0.015	8.17s
	Both	-0.012 ± 0.180	0.018 ± 0.035	10.95s

Table 2: Louisiana bias and execution time for each model.

Method	Model	$\beta_{1,1}$ bounds		$\beta_{2,1}$ bounds		Local Error		Global Error	
		Size	Acc.	Size	Acc.	MAE	MSE	MAE	MSE
Covariate model	No covariate	0.024	69%	0.000	0%	0.028	0.003	0.028	0.014
	Income	0.025	70%	0.001	8%	0.028	0.003	0.028	0.014
	Education	0.024	69%	0.000	0%	0.028	0.003	0.028	0.014
	Both	0.025	73%	0.003	9%	0.028	0.003	0.028	0.014
Spatial model	No map	0.031	100%	0.049	97%	0.021	0.002	0.021	0.011
	Louisiana	0.028	95%	0.035	81%	0.025	0.003	0.025	0.012
Cluster model	1 cluster	0.002	0%	0.000	0%	0.105	0.020	0.039	0.020
	2 clusters	0.002	2%	0.000	0%	0.081	0.014	0.035	0.018
	3 clusters	0.003	2%	0.000	2%	0.059	0.007	0.030	0.015
	4 clusters	0.007	6%	0.015	8%	0.058	0.010	0.006	0.003
Latent structure model	ecolRxC	0.026	62%	0.063	62%	0.031	0.004	0.024	0.012
Linear programming model	nsIphom	0.387	100%	1.000	100%	0.027	0.003	0.027	0.014
Our model	No covariate	0.072	100%	0.158	100%	0.011	0.001	0.011	0.005
	Income	0.075	100%	0.153	100%	0.020	0.002	0.020	0.010
	Education	0.077	100%	0.162	100%	0.020	0.002	0.020	0.010
	Both	0.109	100%	0.258	100%	0.019	0.004	0.003	0.002
Our model without random effects	No covariate	0.002	0%	0.000	0%	0.105	0.020	0.039	0.020
	Income	0.003	2%	0.001	0%	0.102	0.018	0.029	0.015
	Education	0.003	3%	0.000	0%	0.110	0.018	0.036	0.018
	Both	0.004	2%	0.001	0%	0.105	0.016	0.027	0.014

Table 3: Louisiana bounds and error for each model.

Model	$Y_{1,1,1}$	$Y_{1,2,1}$	$Y_{1,1,2}$	$Y_{1,2,2}$
True counts	6878	3376	12504	114
Covariate model : Both	6763	3490	12619	0
Spatial model : No map	6791	3473	12591	18
Cluster model : 4 clusters	6977	3490	12405	0
Latent structure model : ecolRxC	6811	3443	12571	47
Linear programming model : nslphom	6767	3487	12615	3
Our model : No covariate	6830	3447	12552	43
Our model without random effects : Both	5689	3485	13693	5

Table 4: Counts predicted for Acadia for the best models of each method

due to an inherent advantage, caused by the model’s prior structure or a tendency to favour a particular solution well-aligned with this specific dataset, rather than a general improvement in estimation accuracy. To investigate these issues more systematically, we employ artificial datasets derived from real election data structures. By generating synthetic local transition matrices over known aggregate marginals, we can simulate a range of plausible individual-level behaviours while preserving realistic electoral patterns. This controlled setting enables us to critically assess model performance under varying conditions and to better understand the effects of prior assumptions, thereby offering a more robust framework for validating ecological inference methods.

The election data used is the New Zealand votes for the different electorates (a subdivision of New Zealand to guarantee roughly equal population for the elections). In these elections, the people vote for both, a candidate (unique to each electorate) and a party. Then the parliament is filled in part with the chosen candidates, and the rest according to the proportion of votes each party got. As the vote for candidate and party are done at the same time (in fact in the same ballot), the crossed table is known but only the marginals and global transition table are published. For ease of use and because it ensures congruency in the quantities, we used the preprocessed data available in the `ei.Datasets` R package[20] (<https://cran.r-project.org/package=ecolRxC>) As the candidates are different between electorates, each electorate must be treated as a distinct ecological inference problem.

Additionally, to ensure that the specific dataset is not favouring one method over the other, we use the data of two different election results. The first one is the results of Auckland Central of the general electorate, and the second one is the results of Waiariki, of the Māori electorate. The decision of these two electorates was made such that they would not share either land, the two electorates are located in different parts of the north island; nor population, the general electorate is open for anyone (with New Zealand nationality), while the Māori electorate, as its name implies, is only for Māori descendants.

5.2.1 Generation Method

To create realistic yet controlled artificial data for evaluating the ecological inference models, we base our generation process on reference elections for which we know the global transition matrix ($\beta^{(G)}$). These elections provide the empirical structure of real-world scenarios while allowing us to inject synthetic variability at the region level in a principled way. As our focus lies in evaluating the two covariate models, King's and our own, that utilize the multinomial-Dirichlet distribution, we adopt the same framework for generating synthetic data, ensuring consistency between data generation and model specification.

The two key parameters we'd like to control in the behaviour of the synthetic data are:

- **Explainability:** This governs the extent to which a covariate can explain the variation in voting behaviour. Operationally, we define it as the expected absolute correlation between the covariate $\vec{X}_{1:T}$, and the latent transition probabilities $\vec{\beta}_{1:T, r, c}$:

$$\text{explainability} = \mathbb{E} \left[\left| \text{Corr}(\vec{X}_{1:T}, \vec{\beta}_{1:T, r, c}) \right| \right]$$

This means that higher values imply stronger covariate effects. Although we should define the correlations with the logit transformations of the transition probabilities, due to limitations with the JAGS sampler for the MCMC algorithm, we approximate it with the untransformed version.

- **Noise:** This controls the overall variance across regions. Formally, we define noise as a parameter inversely related to the Dirichlet concentration parameter λ with an adjustment ($\lambda = \frac{1}{\text{noise}} - 1$). This allows us to map the interval of noise $(0, 1)$ to the range of variance exhibited by the distribution. To then find the noise of the generated data, we can use the definition of the variance in the Dirichlet distribution to retrieve it:

$$\begin{aligned} \text{Given } \begin{cases} \vec{\beta} \sim \text{Dirichlet}(\lambda \cdot \vec{a}) \\ \sum_i a_i = 1 \end{cases} \\ \text{Var}(\beta_i) = \frac{\mathbb{E}[\beta_i] \cdot (1 - \mathbb{E}[\beta_i])}{\lambda + 1} \implies \frac{\text{Var}(\beta_i)}{\mathbb{E}[\beta_i] \cdot (1 - \mathbb{E}[\beta_i])} = \frac{1}{\lambda + 1} \\ \lambda = \frac{1}{\text{noise}} - 1 \implies \text{noise} = \frac{1}{\lambda + 1} = \frac{\text{Var}(\beta_i)}{\mathbb{E}[\beta_i] \cdot (1 - \mathbb{E}[\beta_i])} \end{aligned}$$

Then, to combine all estimations of the noise, the mean can be used. Higher noise values result in greater dispersion and lower confidence in the transitions, reflecting more region-specific randomness.

This generative framework enables us to systematically explore a range of ecological inference scenarios: from highly explainable and low-noise settings; to noisy, weakly-structured ones; while remaining anchored to realistic election data structures. The goal is to create a controlled yet plausible environment in which to assess the strengths and limitations of our

modelling strategy against its predecessor. The expected effect of these two parameters in the inference is the following. For the noise (how much heterogeneity the data exhibits) we predict a positive correlation with the error of the model estimates. As for the explainability, it should make the inclusion of the covariate better for the model's estimate. Starting from an almost negligible difference for variables with low correlation up to around half the error of the model for the ones with the highest tested correlation (about 60% ~ 65%).

As the model to generate the data assumes the distribution of King and ours model, the first two stages will be mostly the same. The total counts are modelled with a multinomial, its probabilities as the application of the transition matrix and the transition matrix with a Dirichlet. The changes made are first the parameters of said Dirichlet and the additions to coerce the model to one with the explainability and noise we want. For the parameters of the Dirichlet, the known global transition matrix is used for the expected value, and the desired level noise for the variance of the distribution. Meanwhile, for the two key parameters, the measure is calculated for the current estimation and a penalization is added depending on how far from the desired value it is. To quickly add this penalization without too many complications, a beta distribution is used with mean equal to the desired value. A "force" parameter is added to both to manage how tightly this sampled value adheres to the expected, allowing for flexible control over how deterministic the covariate's effect is, and how much heterogeneity appears in the data. Putting this all together produces model 9.

5.2.2 Results

Our empirical results largely confirm what we expected, though with some notable nuances. As predicted, higher noise levels leads to increased error rates for all models in both datasets (figures 2, 3, 4 and 5). Similarly, the presence of a more tightly correlated explanatory variable did widen the performance gap between covariate-informed and covariate-agnostic models. However, the magnitude of this gap is less pronounced than we initially were anticipating, especially in scenarios with moderate levels of explainability (under 50%), where barely any change can be seen. This indicates that only strongly correlated covariates can meaningfully contribute to reducing estimation error, while the inclusion of weakly correlated variables will offer little benefit. In the case of a covariate with very low correlation, its addition can even worsen the result of the global transition matrix as can be best seen in figure 5, for the Māori electorate.

Across all tested scenarios, our proposed model consistently matched or outperformed the original covariate model proposed by King, Rosen, Tanner and Jiang. Notably, in multiple cases where their model showed great improvement when covariates were added, our model had already achieved the lower error levels even in the absence of those enhancements. This performance advantage held across both local (for all region tables) and global (aggregated table) error rates.

In addition to precision, computational efficiency displayed a clear difference between both models. Our model achieved these results while requiring significantly less runtime, typically between one-half and one-fifth of the time required by the original covariate model (figures 6 and 7). Moreover, our implementation demonstrated robust numerical stability; it successfully completed all experimental runs without failure. In contrast, the model King, Rosen, Tanner

Data transformation:

$$p_{t,r}^{(1)} = Y_{t,r}^{(1)} / N_t^{(1)} \quad \forall t \in \{1, \dots, T\}, r \in \{1, \dots, R\}$$

First stage:

$$\begin{aligned} \vec{Y}_{t,1:C}^{(2)} &\sim \text{Multinomial}(N_t^{(2)}, \vec{\theta}_{t,1:C}^{(2)}) & \forall t \in \{1, \dots, T\} \\ \theta_{t,c}^{(2)} &= \vec{p}_{t,1:R}^{(1)} \cdot \vec{\beta}_{t,1:R,c} & \forall t \in \{1, \dots, T\}, c \in \{1, \dots, C\} \end{aligned}$$

Second stage:

$$\begin{aligned} \vec{\beta}_{t,r,1:C} &\sim \text{Dirichlet}(\lambda \cdot \vec{b}_{r,1:C}^{(G)}) \\ \lambda &= \frac{1}{\text{noise}} - 1 \\ \text{corr.} &= \left| \text{Corr}(\vec{X}_{1:T}, \vec{\beta}_{1:T,r,c}) \right| & \forall r \in \{1, \dots, R\}, c \in \{1, \dots, C\} \\ \text{coef.} &= \frac{\text{Var}(\vec{\beta}_{1:T,r,c})}{\mathbb{E}(\vec{\beta}_{1:T,r,c}) \cdot (1 - \mathbb{E}(\vec{\beta}_{1:T,r,c}))} & \forall r \in \{1, \dots, R\}, c \in \{1, \dots, C\} \end{aligned}$$

Third stage:

$$\begin{aligned} \text{explainability} &\sim \text{Beta}(\text{force}^{(\text{exp.})} \cdot \mathbb{E}[\text{corr.}], \text{force}^{(\text{exp.})} \cdot (1 - \mathbb{E}[\text{corr.}])) \\ \text{noise} &\sim \text{Beta}(\text{force}^{(\text{noise})} \cdot \mathbb{E}[\text{coef.}], \text{force}^{(\text{noise})} \cdot (1 - \mathbb{E}[\text{coef.}])) \end{aligned}$$

Model 9: Generation of a plausible scenario, defined in section 5.2.1.

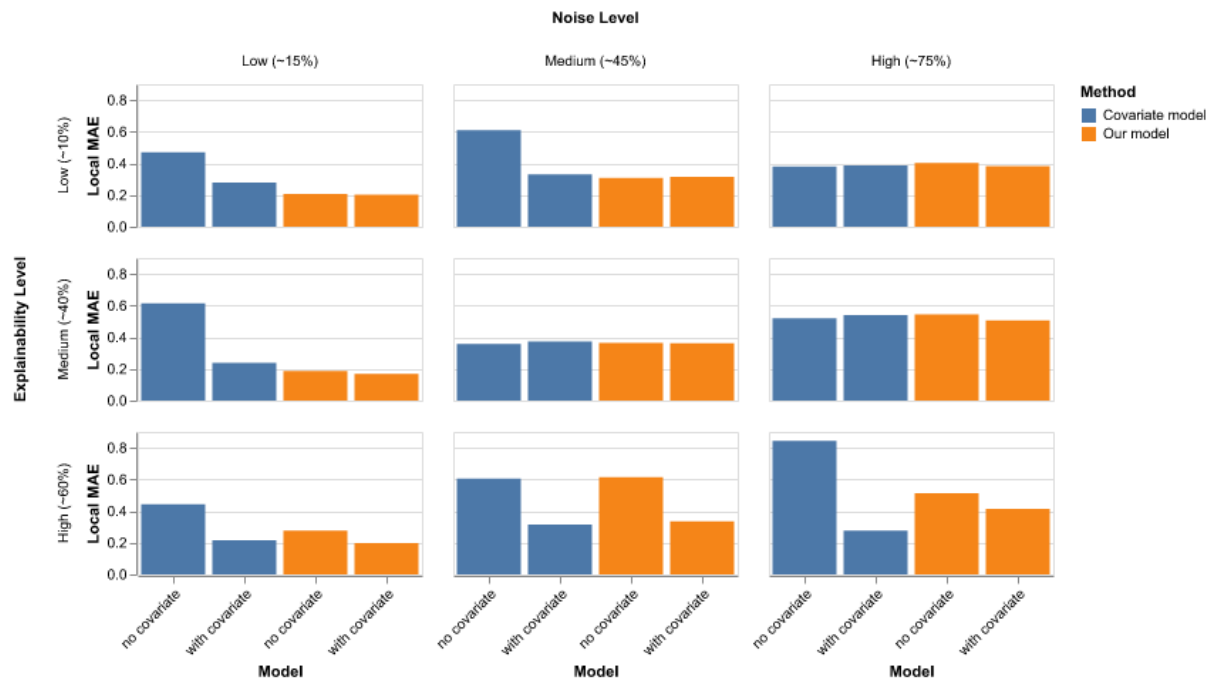


Figure 2: Local mean absolute error for the Auckland Central general electorate.

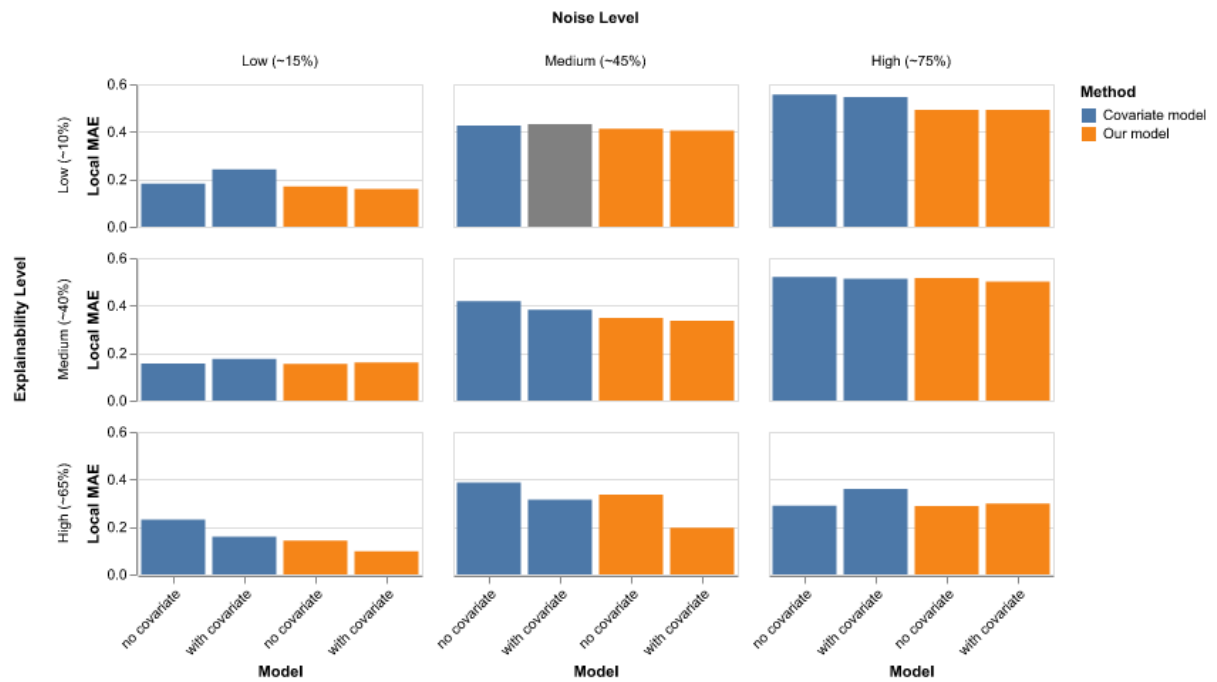


Figure 3: Local mean absolute error for the Waiariki Māori electorate. (Numerical approximation in gray)

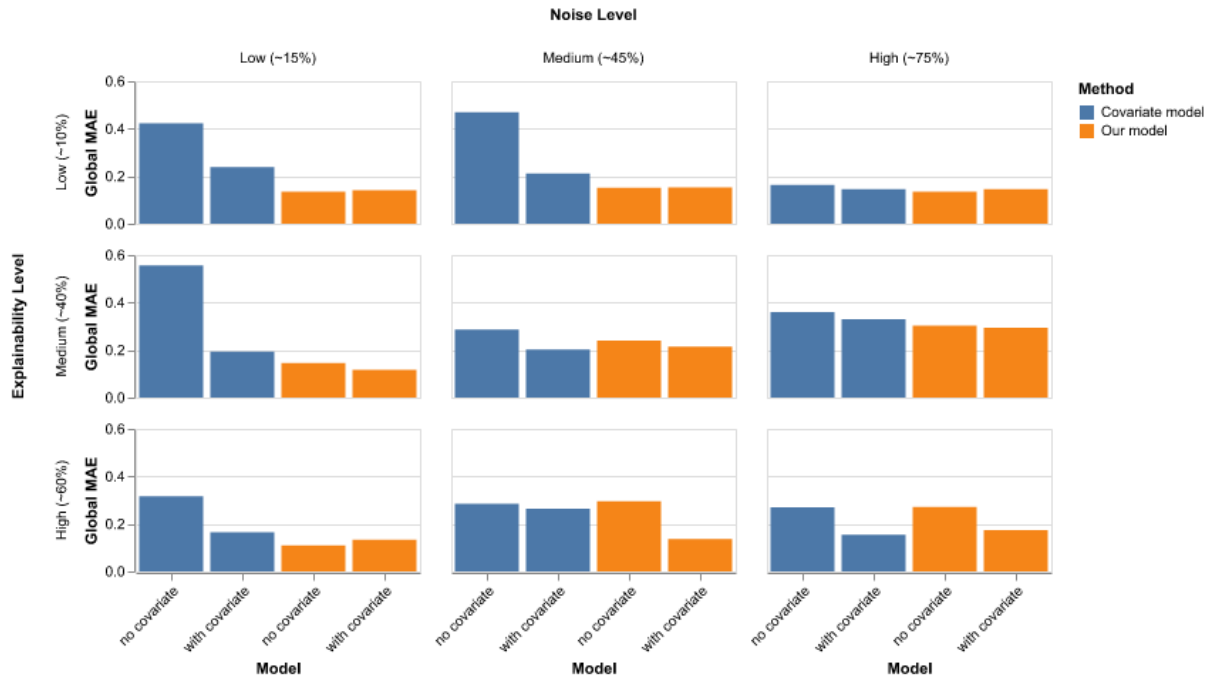


Figure 4: Global mean absolute error for the Auckland Central general electorate.

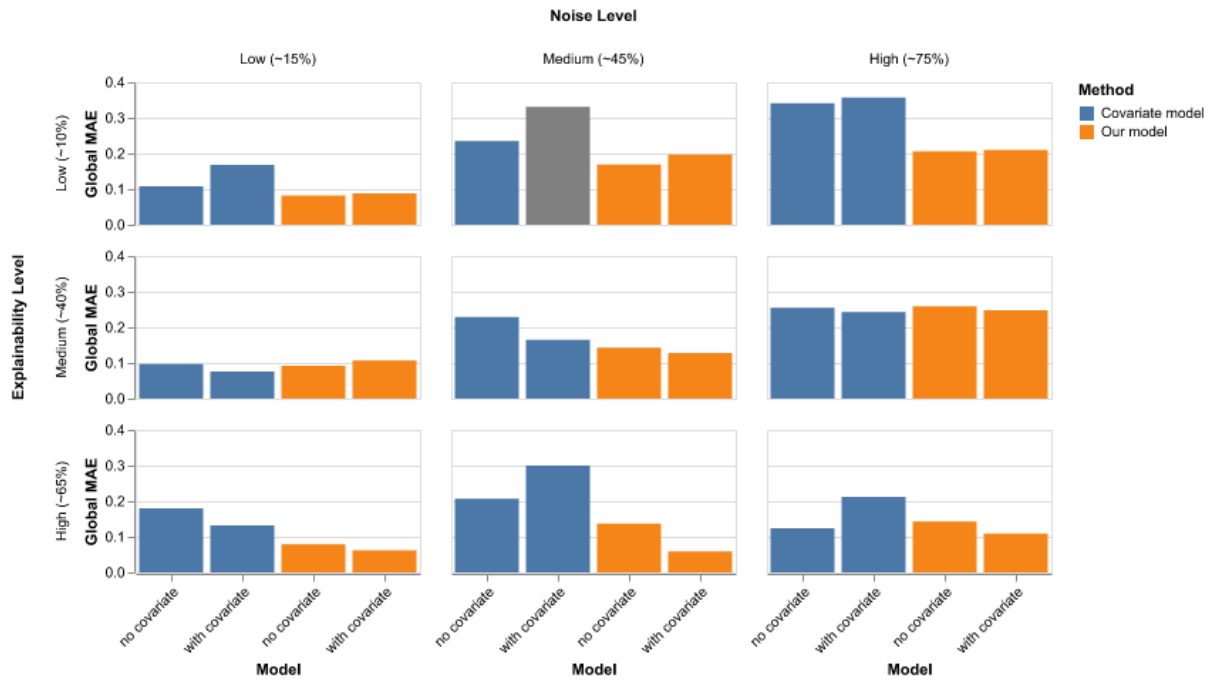


Figure 5: Global mean absolute error for the Waiariki Māori electorate. (Numerical approximation in gray)

and Jiang proposed encountered numerical issues due to values too extreme (infinities, zero probability, etc.), leading to incomplete estimates, replaced with the numerical approach described in the same paper[11]. Even though we tried to remedy the numerical issues on King's model; by normalizing the input values, adding limits to the unbound distributions and capping the exponential result; some of the runs still were unable to run successfully.

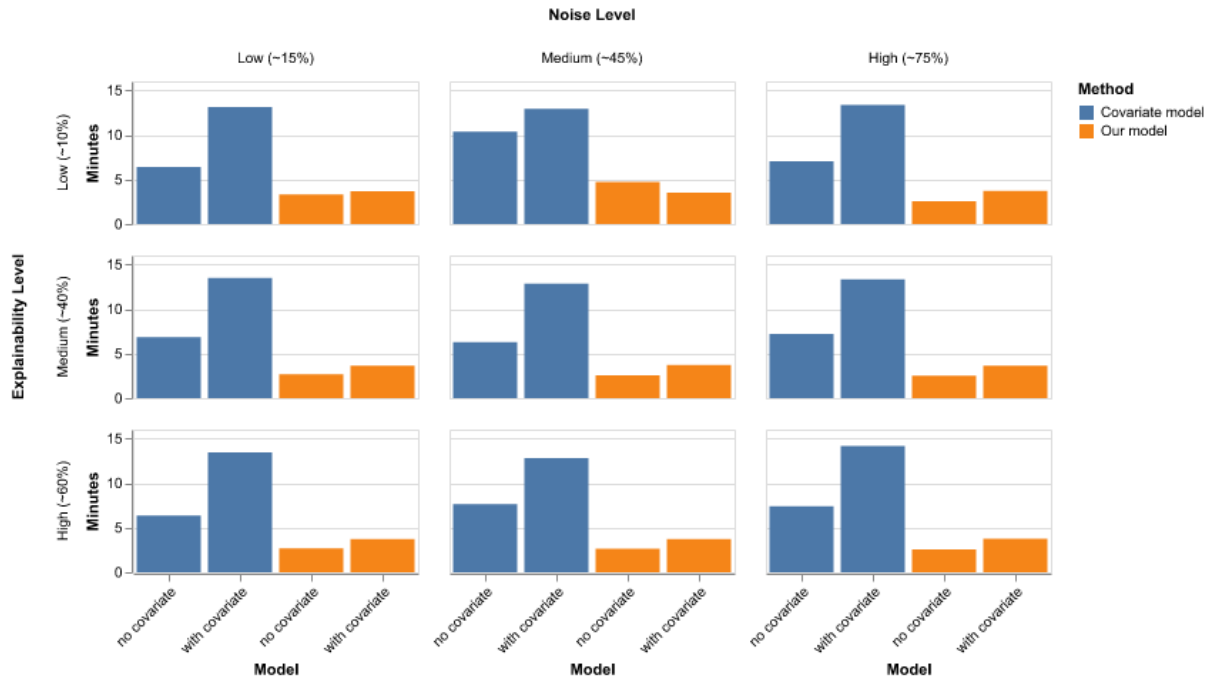


Figure 6: Execution time in minutes for the Auckland Central general electorate.

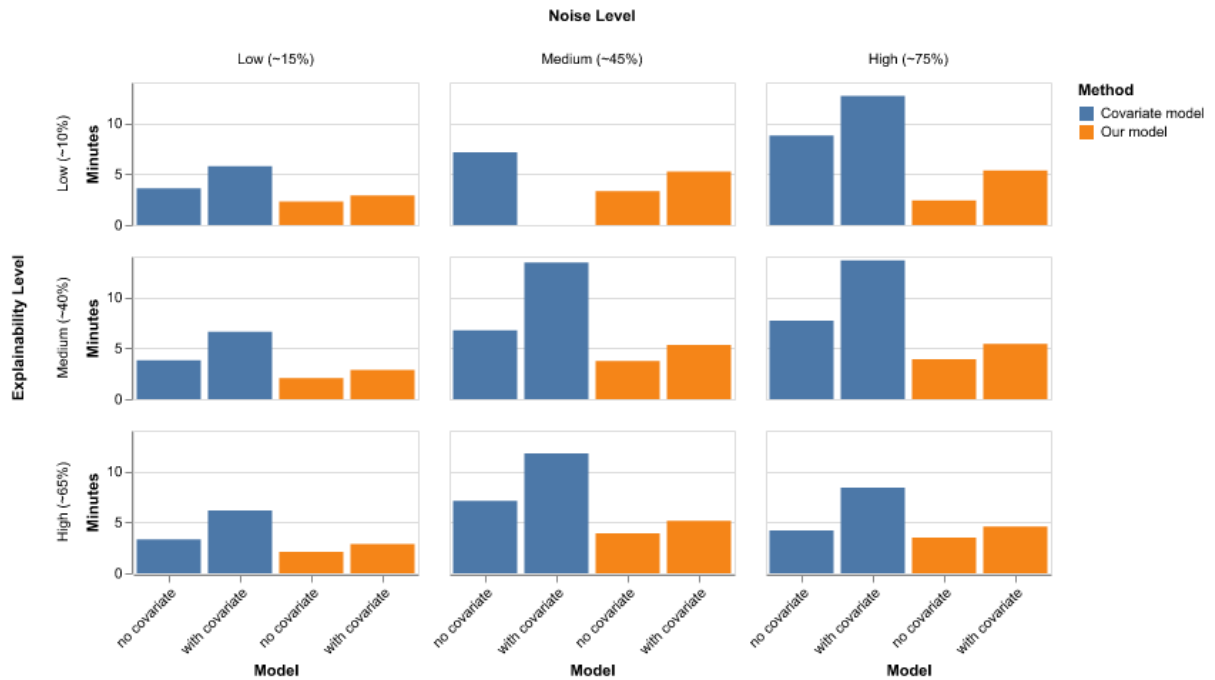


Figure 7: Execution time in minutes for the Waiariki Māori electorate. (Empty for failed runs)

6 Conclusion

In this paper, we proposed a refined ecological inference model that extends the multinomial-Dirichlet framework introduced by King, Rosen, Tanner, and Jiang[11]. Our formulation reconsiders the way covariates are represented and modelled, and offers an alternative pathway for incorporating them into the model. This approach retains the original framework’s flexibility, but addresses some of its practical limitations in computational efficiency and model stability. In addition, we assessed the usefulness of the inclusion or exclusion of random effects.

We evaluated the proposed model using both real-world voter registration data from Louisiana and a suite of controlled simulated scenarios. For all the datasets, our model consistently matched or outperformed the original covariate approach. Importantly, it achieved this while being substantially faster, often requiring between half and one-fifth of the computational time, of its predecessors and demonstrating superior numerical stability. Unlike the original model, which occasionally failed to run the MCMC algorithm, our model ran reliably in all experiments.

We also found that excluding random effects can yield substantial speed-ups, often by an order of magnitude, without sacrificing much accuracy if working at the global level. This suggests that in some applications, especially those with major focus on the overall transition or its decomposition into clusters or correlation, these models may be preferable.

These evaluations confirmed that data with greater heterogeneity lead to higher estimation error. Additionally, we could see that not all covariates meaningfully contribute only if they are strongly correlated (above 40%) with the latent structure. However, this also means that covariates with low correlation to the transition matrix have little to no impact on model performance, reinforcing the need for careful covariate selection. On top of that, we saw that the faster numerical models (nslphom and ecolRxC) achieved results similar to more complex Bayesian approaches but at a fraction of the runtime. This reinforces a key insight: model complexity must be justified, (for example having variable very likely tied to the transition of votes) otherwise simpler approaches may be preferable. Together, these findings contribute to a clearer understanding of when model complexity is warranted and when streamlined approaches are sufficient.

Looking ahead, several avenues for future work emerge. One promising direction is the creation of numerical approximation schemes, such as variational inference or moment-based methods, to further reduce runtime without sacrificing much accuracy. Another is the development of a more thought-out and tested prior for the covariate parameters, which could improve estimations and stabilize estimation in scenarios with a less informative covariate. Finally, while this study focused on electoral data, the model’s flexibility makes it well-suited for applications in other domains that rely on aggregate data, such as epidemiology, education, or market research.

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